

FS Migration Validation Engine

Microsoft Founders Hub Application Pack

Technical Master Document

Version 2.0 — June 2026

Status: Ready for Review

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Phase 1 – Azure Founders Hub Pack (v4 Cloud-Ready Edition)

Version: 2.0

Based on: FS Migration Validation Engine v1.4

Target Program: Microsoft for Startups Founders Hub

Status: Ready for Review

Executive Summary

This pack presents the **FS Migration Validation Engine** — an automated, metadata-driven data migration validation platform purpose-built for regulated Financial Services institutions.

After extensive technical analysis (Parts 1–4) covering legacy system assessment, current architecture evaluation, gap analysis, and migration strategy, we have transitioned from research into productization. This documentation pack represents the complete technical, commercial, and strategic foundation for Microsoft Founders Hub application.

Product Positioning

An Automated Data Migration Validation & Governance Platform for Regulated Financial Services.

The platform is no longer just a data migration tool. It has evolved into an enterprise-grade Migration Assurance Platform that automates legacy system discovery, intelligent schema mapping, migration validation, governance, scoring, and compliance auditing for regulated financial data migrations.

The Five Platform Pillars

Pillar	Description	Status
1. Enterprise Discovery	Automated schema discovery, metadata extraction, dependency analysis	[OK] Built
2. Intelligent Mapping	Schema comparison, similarity matching, confidence scoring	[OK] Built
3. Migration Validation	10 structured controls, rule execution, exception tracking	[OK] Built
4. Governance & Audit	Version control, audit history, compliance reporting, release gates	[OK] Built
5. AI Intelligence Layer	AI-assisted mapping recommendations, analytics,	 In Development

	copilot (future)	
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Document Inventory

#	Document	Audience	Purpose
1.1	Product Overview	Microsoft reviewers, investors	Executive summary, problem/solution, value proposition
1.2	Technical Architecture	Architecture reviewers, engineering	Platform architecture, components, data model
1.3	Platform Core Definition	Architecture reviewers	Orchestration layer, event bus, plugin framework
1.4	Azure Cloud Architecture	Microsoft reviewers, cloud architects	Azure service mapping, deployment, Well-Architected alignment
1.5	Azure Reference Architecture Diagram	Technical reviewers	Visual architecture diagram, scalability, security boundaries
1.6	Technical Narrative	Founders Hub, investors	Full narrative: problem, solution, why Azure, roadmap

Evidence Portfolio

All claims in this pack are backed by working code and technical analysis:

Evidence	Reference	Status
[OK] Legacy Assessment	Part 1 Analysis	Complete
[OK] Current Architecture Assessment	Part 3 Analysis	Complete
[OK] Migration Strategy	Part 4 Analysis	Complete
[OK] Working CLI Engine	`app/main.py` — 10 controls, scoring, audit	Complete

[OK] FastAPI Endpoints	`app/api/` — REST API layer	Complete
[OK] Database Schema	`sql/schema/` — PostgreSQL engine schema	Complete
[OK] Docker Deployment	`Dockerfile`, `docker-compose.yml`	Complete
[OK] Governance Framework	Release gates, control dependencies, audit export	Complete
[OK] Gap Analysis	Phase 1.3 NOT_REQUIRED doc	Complete
[OK] Security Scanning	`bandit-report.html`, `trivy-fs-report.txt`	Complete

Target Market

Primary: Financial Services — Banks (Tier 1–3), Insurers, Asset Managers, FinTech, Regulated Enterprises

Secondary: Any organisation undertaking regulated data migration (Healthcare, Government, Utilities)

The platform is designed for organisations where data integrity, compliance, and auditability during migration are non-negotiable requirements.

Program Alignment

This documentation pack is designed for:

- [OK] **Microsoft for Startups Founders Hub** — Primary target
- [OK] **AWS Activate** — Adaptable with minor cloud service mapping changes
- [OK] **Google for Startups** — Adaptable with minor cloud service mapping changes
- [OK] **Investor Due Diligence** — Angel, Seed, Series A
- [OK] **Enterprise Customer Evaluations** — Technical and commercial reviews

Key Differentiators

Feature	Our Platform	Manual Approach	Legacy ETL Tools
Automated schema discovery	[OK] Automated	[X] Manual analysis	[X] Requires pre-configuration
10 structured validation	[OK] Built-in	[X] Ad-hoc testing	[!] Partial

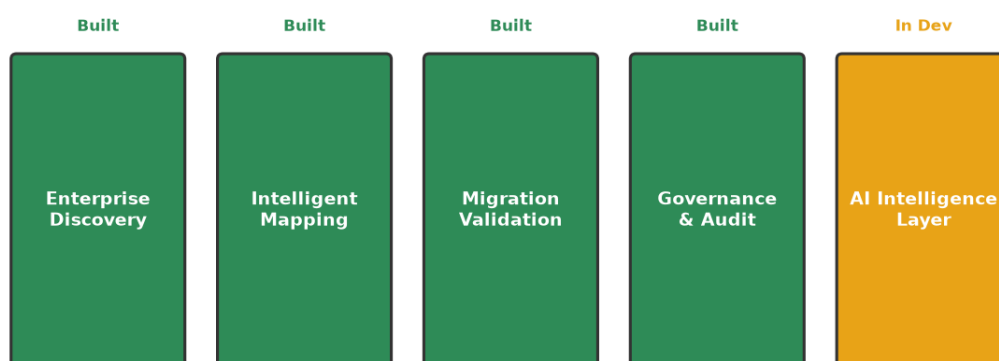
controls			
Confidence scoring	[OK] Automated	[X] Subjective	[X] Not available
Audit trail & governance	[OK] Built-in	[X] Manual documentation	[!] Limited
Release gate enforcement	[OK] Automated	[X] Manual approval	[X] Not available
Control dependency management	[OK] Built-in	[X] Manual sequencing	[X] Not available
Docker deployment	[OK] Ready	N/A	[!] Legacy installs
Financial Services focus	[OK] Purpose-built	[X] Generic	[X] Generic

Next Steps

Following this master overview, review the documents in order:

1. **Phase 1.1** — Product Overview (non-technical executive summary)
2. **Phase 1.2** — Technical Architecture (engineering deep-dive)
3. **Phase 1.3** — Platform Core Definition (orchestration architecture)
4. **Phase 1.4** — Azure Cloud Architecture (cloud-native migration plan)
5. **Phase 1.5** — Azure Reference Architecture Diagram (visual architecture)
6. **Phase 1.6** — Technical Narrative (complete Founders Hub narrative)
7. **Phase 2.1–2.10** — Commercial & Investor Readiness Package

Five Platform Pillars



Five Pillars

Phase 1.1 — Product Overview

Version: 2.0

Product: FS Migration Validation Engine

Target: Microsoft for Startups Founders Hub

Audience: Non-technical reviewers, investors, enterprise customers

Executive Summary

Financial Services institutions undertaking data migration projects face a fundamental challenge: how to ensure that millions of records move from legacy systems to modern platforms **accurately, completely, and in compliance with regulatory requirements**.

Traditional approaches rely on manual sampling, ad-hoc spreadsheet tracking, and post-migration firefighting. This is slow, error-prone, and unacceptable in regulated environments where data integrity failures can trigger regulatory penalties, operational losses, and reputational damage.

The FS Migration Validation Engine solves this by providing an **automated, metadata-driven validation platform** that systematically verifies every aspect of a data migration — from schema compatibility and data type correctness through to referential integrity, business rule compliance, and regulatory reporting accuracy.

The Problem

Data migration in Financial Services is high-risk and highly regulated:

Challenge	Impact
Manual validation	~60% of validation effort is manual and subjective
Incomplete coverage	Sampling-based checks miss edge cases and exceptions
Poor auditability	Spreadsheet-driven tracking fails regulatory scrutiny
No repeatability	Every migration project starts from scratch
High cost of failure	Data errors can cause operational losses, regulatory fines
Lack of governance	No formal release gates, approval workflows, or sign-off trails

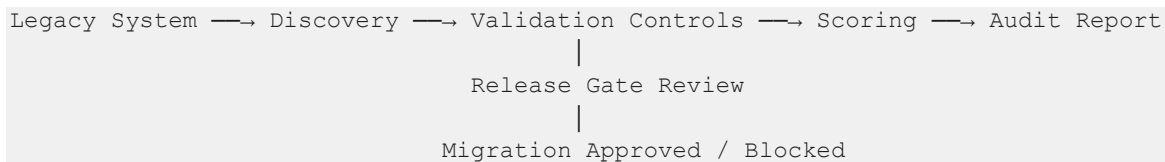
Banks migrating core banking systems, payment platforms, or regulatory reporting systems cannot afford data quality failures. Yet most migration validation today relies on custom scripts, manual checks, and hope.

Our Solution

The FS Migration Validation Engine is an **automated, metadata-driven validation and governance platform** that:

8. **Discovers** legacy and target system schemas automatically
9. **Validates** data across 10 structured migration controls
10. **Scores** migration quality with objective, repeatable metrics
11. **Governs** the migration lifecycle with release gates and audit trails
12. **Reports** comprehensive audit evidence suitable for regulatory review

How It Works



The platform runs as a CLI engine, REST API, or Docker container — integrating into existing CI/CD pipelines and deployment workflows.

Platform Capabilities

1. Automated Discovery

- Connect to source and target databases
- Discover schemas, tables, columns, and relationships
- Extract metadata for validation planning
- Identify schema differences automatically

2. Structured Validation Controls (C01–C010)

Control	Description
C01	Row count reconciliation
C02	Column-level data type validation
C03	Nullability constraint verification

C04	Primary key integrity check
C05	Foreign key referential integrity
C06	Business rule validation
C07	Date/time boundary validation
C08	Numeric range and precision checks
C09	String pattern and format validation
C010	Regulatory field completeness check

3. Scoring & Release Gates

- Each control receives an objective pass/fail score
- Aggregate scoring provides migration quality metrics
- Configurable release gates block migrations below minimum quality thresholds
- Dependency management ensures controls execute in correct order

4. Governance & Audit

- Every control execution is logged with timestamp and result
- Audit-ready CSV exports for regulatory review
- Batch-level tracking with unique batch IDs
- Exception register for all validation failures
- Full execution history for compliance reporting

5. REST API

- FastAPI-based RESTful interface
- Run validations programmatically
- Integrate with CI/CD pipelines
- Query results and export audits remotely

Business Benefits

Benefit	Measurable Impact
Reduced manual effort	70-80% reduction in manual validation work
Complete coverage	100% of records validated, not just samples
Regulatory compliance	Audit-ready evidence for every migration

Repeatable process	Standardised validation across all migration projects
Faster migrations	Automated gates replace manual review cycles
Lower risk	Early detection of data quality issues
Governance by design	Release gates prevent bad data reaching production

Current Development Status

Component	Status
CLI Engine (10 controls, scoring, audit)	[OK] Production-ready
REST API layer	[OK] Built (FastAPI)
Database schema (PostgreSQL)	[OK] Production-ready
Docker deployment	[OK] Ready
Control dependency DAG	[OK] Implemented
Release gate enforcement	[OK] Implemented
Audit export (CSV)	[OK] Implemented
Security scanning (bandit, trivy)	[OK] Integrated
Web dashboard	 In development
AI-assisted mapping recommendations	 Planned
Multi-tenant SaaS deployment	 Planned

Why Azure

Microsoft Azure provides the enterprise-grade foundation required for regulated Financial Services deployments:

- **Azure Container Apps** — Scalable, serverless compute for the validation engine
- **Azure SQL Database** — Managed relational database with high availability
- **Azure Blob Storage** — Secure audit report and evidence storage
- **Azure Key Vault** — Encrypted secrets management for database credentials
- **Microsoft Entra ID** — Enterprise identity and role-based access control

- **Azure Monitor** — Centralised logging and performance monitoring
- **Azure API Management** — Secure API gateway for enterprise integration

The platform is designed to be Azure-native while supporting hybrid and on-premises deployment for customers with specific regulatory requirements.

Product Vision

Our vision is to become the **standard platform for regulated data migration assurance** — providing Financial Services institutions with the automation, governance, and auditability they need to migrate mission-critical data with confidence.

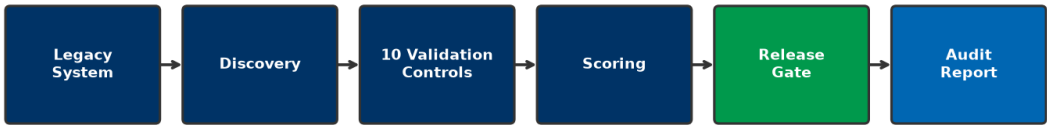
Future capabilities include:

- **AI-assisted mapping recommendations** — Using schema analysis to suggest optimal field mappings
- **Intelligent mapping engine** — Automated schema matching with confidence scoring
- **Interactive dashboards** — Real-time migration quality visualisation
- **Multi-cloud support** — Extend validation to AWS, GCP, and hybrid deployments
- **Regulatory template library** — Pre-built controls for specific regulations (Basel, SOX, FCA, PRA)

Competitive Advantages

Area	Our Advantage
Purpose-built	Designed specifically for regulated Financial Services data migration
10 structured controls	Comprehensive validation framework, not a generic ETL tool
Governance-first	Release gates, audit trails, and compliance built into every execution
Automated scoring	Objective, repeatable quality metrics — not subjective assessments
Docker-native	Deploy anywhere — cloud, on-premises, hybrid
API-first	Integrate with existing CI/CD and deployment pipelines
Open, extensible	Python-based, open architecture, easily customisable controls

How It Works — Automated Validation Workflow



How It Works

Phase 1.2 — Technical Architecture

Version: 2.0

Product: FS Migration Validation Engine

Status: Architecture Baseline

Audience: Microsoft Founders Hub, Technical Reviewers, Engineering Team

1. Introduction

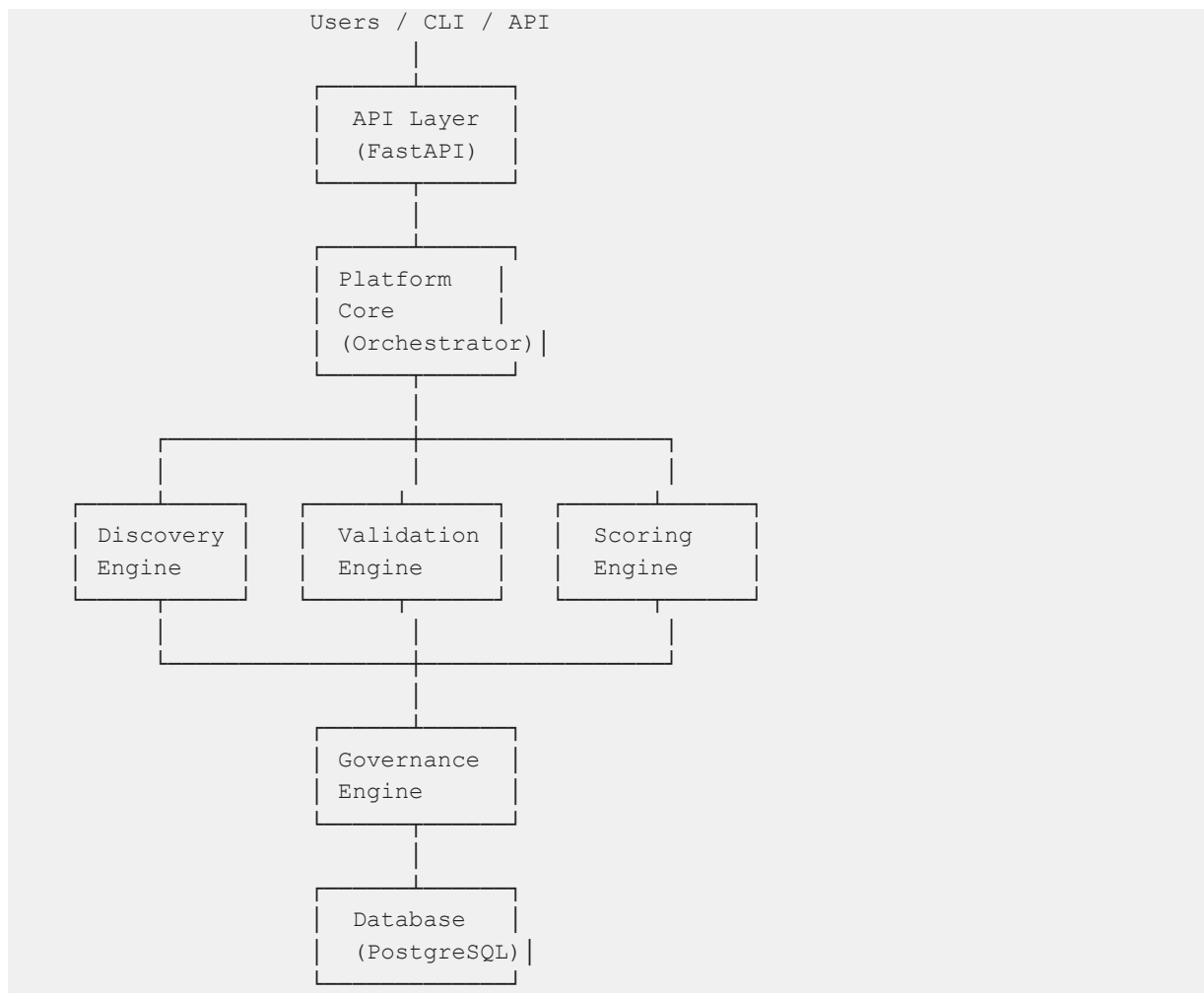
The FS Migration Validation Engine is an automated, metadata-driven data migration validation and governance platform designed for regulated Financial Services environments. It validates data migrations across core banking, payments, lending, asset management, and regulatory reporting systems.

The platform is architected as a modular, cloud-ready solution with a centralised orchestration layer (Platform Core) that coordinates independent domain engines responsible for discovery, validation, scoring, and governance.

2. Architectural Principles

Principle	Description
Modularity	Independent engines communicate through the Platform Core
Metadata-driven	All validation is driven by database schema metadata
Security-by-design	Zero Trust, encryption at rest and in transit, secrets isolation
Governance-first	Release gates, audit trails, and compliance built into every execution
Extensibility	Plugin-based engine architecture for adding new controls
Cloud-native	Designed for Azure, deployable anywhere with Docker
Observability	Comprehensive logging, metrics, and audit trails
Repeatability	Deterministic validation — same inputs produce same outputs

3. High-Level Architecture



4. Core Platform Components

4.1 Platform Core

The central orchestration layer responsible for:

- **Workflow orchestration** — Controlling execution sequence of discovery, validation, scoring
- **Control dependency management** — DAG-based execution (C02 depends on C01, C09 depends on C03)
- **State management** — Tracking batch execution state, checkpointing
- **Configuration management** — Centralised config for all engines
- **Security enforcement** — Authentication, authorisation, audit logging
- **Event routing** — Emitting and consuming events between engines
- **Error handling** — Failure isolation, retry logic, graceful degradation

4.2 Discovery Engine

Responsible for automated schema discovery from source and target databases.

Capabilities:

- Database connection management via `app/db_connector.py`
- Schema discovery — tables, columns, data types, constraints
- Metadata extraction and repository population
- Relationship detection (primary keys, foreign keys)

Output: Metadata repository in the engine database.

4.3 Validation Engine

The core validation component executing 10 structured migration controls.

Validation Controls (C01–C010):

Control	Description	SQL Template
C01	Row count reconciliation	`sql/controls/`
C02	Column-level data type validation	`sql/controls/`
C03	Nullability constraint verification	`sql/controls/`
C04	Primary key integrity check	`sql/controls/`
C05	Foreign key referential integrity	`sql/controls/`
C06	Business rule validation	`sql/controls/`
C07	Date/time boundary validation	`sql/controls/`
C08	Numeric range and precision checks	`sql/controls/`
C09	String pattern and format validation	`sql/controls/`
C010	Regulatory field completeness check	`sql/controls/`

Controls are implemented as SQL templates executed against source and target databases, with results stored in the engine database.

4.4 Scoring Engine

Responsible for evaluating validation results and producing quality metrics.

Capabilities:

- Control-level pass/fail determination
- Aggregate scoring across all controls
- Configurable pass thresholds
- Weighted scoring by control criticality
- Exception register management

4.5 Governance Engine

Provides enterprise governance across the migration lifecycle.

Capabilities:

- Release gate enforcement — blocks migration if score below threshold
- Batch-level tracking with unique batch IDs
- Audit trail — every execution logged with timestamp and result
- Exception register — all validation failures recorded
- CSV audit export for regulatory review

5. Data Architecture

5.1 Engine Database (PostgreSQL)

The engine database stores all operational data:

Entity	Description
`migration_validation_batch`	Batch-level execution tracking
`migration_control_execution`	Individual control execution results
`migration_exception_register`	All validation exceptions
Source/target metadata	Discovered schema information

See `sql/schema/01_engine_schema.sql` for full schema definition.

5.2 Source & Target Databases

The platform connects to source and target PostgreSQL databases to perform validation. The connector layer (`app/db_connector.py`) manages connections and handles credential security through environment variables and Azure Key Vault.

6. Security Architecture

Layer	Implementation
Identity	Environment-based authentication, future: Microsoft Entra ID
Secrets	Environment variables → Azure Key Vault migration path
Encryption at rest	PostgreSQL TDE / Azure SQL transparent encryption

Encryption in transit	TLS for all database connections
API security	FastAPI with CORS, future: OAuth2 / Entra ID
Secrets isolation	No credentials in source code — all via environment variables
Audit logging	Every control execution logged with batch ID and timestamp

7. Integration Architecture

7.1 CLI Interface

```
python app/main.py run --config config.yaml
python app/main.py discover --config config.yaml
python app/main.py export --config config.yaml --batch-id <id>
```

7.2 REST API (FastAPI)

- `GET /health` — Health check
- `POST /api/v1/validate` — Run validation batch
- `GET /api/v1/batch/{id}` — Get batch results
- `GET /api/v1/export/{batch_id}` — Export audit CSV

7.3 Docker Deployment

- `docker-compose up --build` — Full stack deployment
- Containerised engine, database, and API

8. Deployment Architecture

Environment	Deployment Model	Database
Development	Local Docker Compose	Local PostgreSQL
Test/QA	Azure Container Apps	Azure SQL Database
Production	Azure Container Apps / AKS	Azure SQL Database
On-premises	Docker Compose	PostgreSQL

9. Current Implementation Status

Component	Status	Location
CLI Engine	[OK] Production-ready	`app/main.py`, `app/execution_engine.py`
10 Validation Controls	[OK] Implemented	`sql/controls/`
Scoring Engine	[OK] Implemented	`app/scoring_engine.py`
Governance / Release Gates	[OK] Implemented	`config.yaml`, `app/execution_engine.py`
Audit Export	[OK] Implemented	`app/audit_export.py`
Database Schema	[OK] Production-ready	`sql/schema/`
Docker Deployment	[OK] Ready	`Dockerfile`, `docker-compose.yml`
REST API	[OK] Built	`app/api/`
Discovery Service	[OK] Built	`app/services/dataset_discovery_service.py`
Rule Executor	[OK] Built	`app/rule_executor.py`, `app/rule_factory.py`
Web Dashboard	 In Development	`dashboard/`
AI Mapping Engine	 Planned	`app/mapping_engine/`
Multi-tenant SaaS	 Planned	Future phase

10. Technology Stack

Layer	Technology
Language	Python 3.11
API Framework	FastAPI, Uvicorn
Database	PostgreSQL 15+
Containerisation	Docker, Docker Compose
Configuration	YAML + Environment Variables
Validation	SQL-driven control templates
Reporting	CSV export (future: interactive dashboards)
Security	python-jose, cryptography

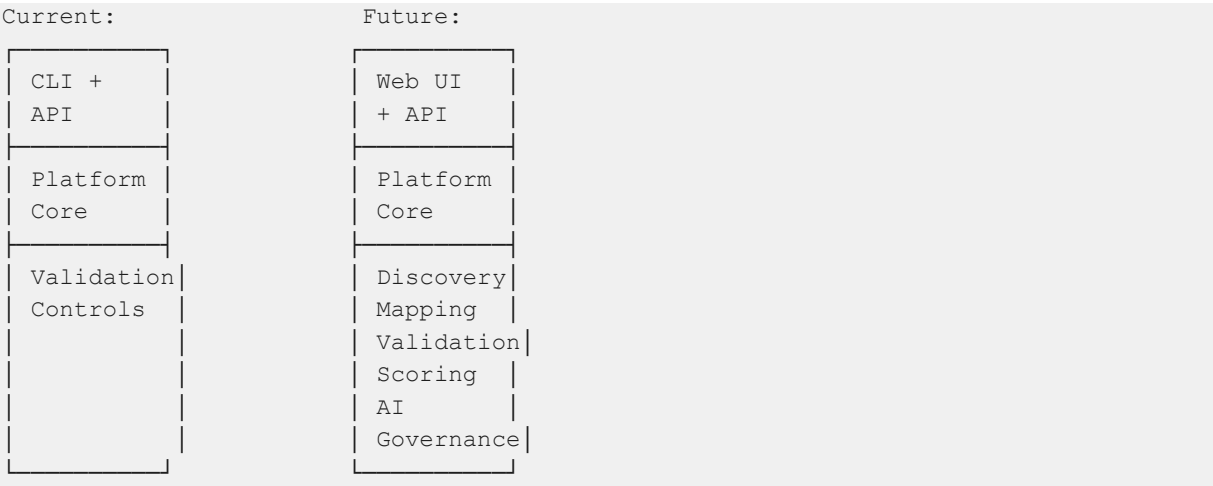
Testing	pytest
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11. Scalability

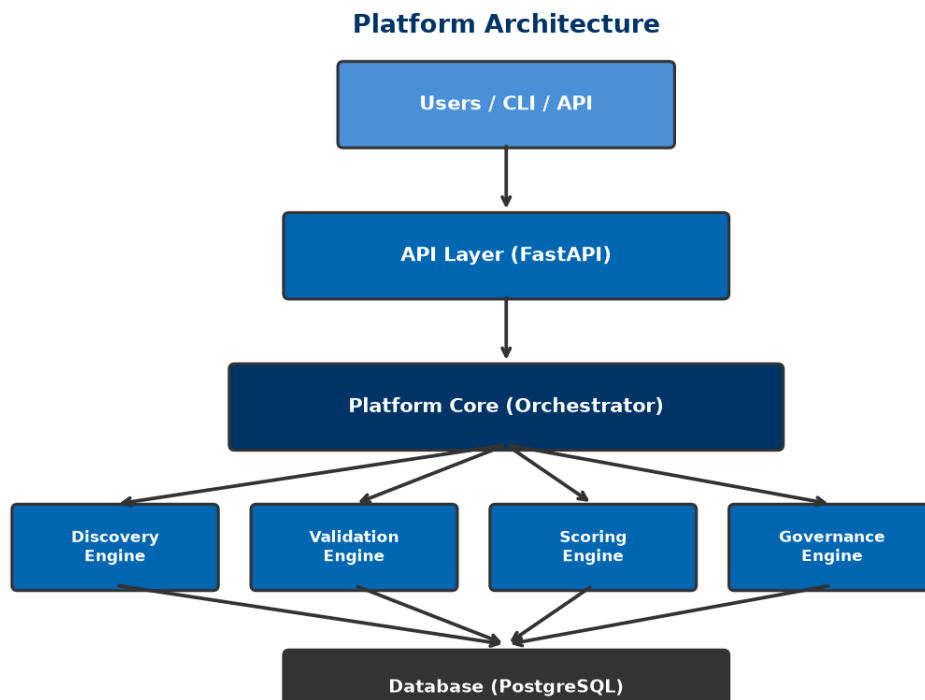
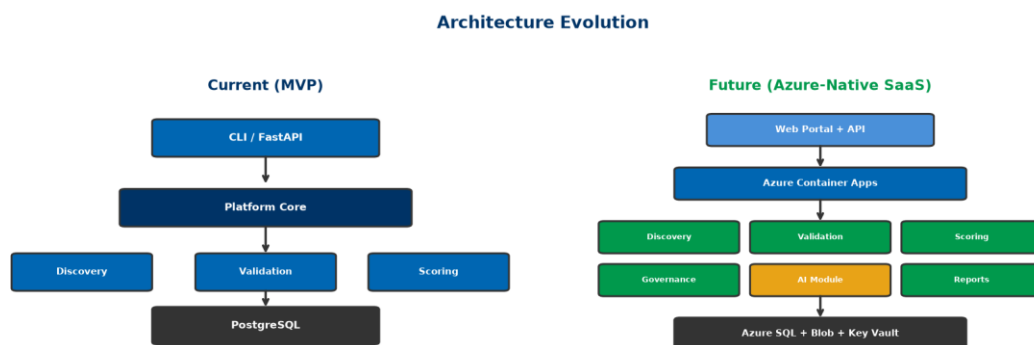
The architecture supports horizontal scaling through:

- **Stateless API** — FastAPI instances can be scaled horizontally
- **Independent engine scaling** — Each engine can scale independently
- **Database connections** — Connection pooling via PostgreSQL
- **Container orchestration** — Kubernetes-ready (Docker images)
- **Async processing** — Future: background task queues for large batches

12. Future Architecture Evolution



The modular Platform Core architecture ensures that new engines can be added without architectural redesign. The AI Intelligence Layer (Azure OpenAI) can be integrated as an additional engine providing mapping recommendations, anomaly detection, and natural language querying.

*Platform Architecture**Architecture Evolution*

Phase 1.3 — Platform Core Definition

Version: 2.0

Status: Architecture Baseline

Audience: Engineering, Architecture, Founders Hub Technical Reviewers

Platform Core

Executive Summary

The **Platform Core** is the central orchestration layer of the FS Migration Validation Engine. Rather than allowing each engine to communicate directly with every other engine, the Platform Core acts as the single coordination point responsible for workflow orchestration, state management, event routing, security enforcement, configuration management, and engine lifecycle management.

This architecture dramatically reduces coupling between components while improving scalability, maintainability, security, and extensibility.

Why the Platform Core Exists

Without a Platform Core:

Discovery ↔ Validation ↔ Scoring ↔ Governance ↔ Export

Every engine develops dependencies on every other engine, creating duplicated logic, inconsistent state, difficult testing, tight coupling, and deployment challenges.

With a Platform Core:



Each engine communicates only with the Platform Core. No engine requires knowledge of another engine's internal implementation.

Core Responsibilities

1. Workflow Orchestration

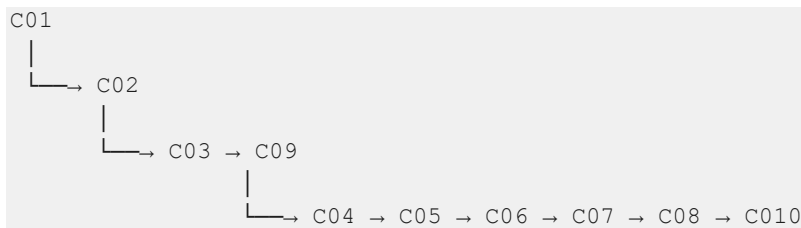
Coordinates execution between engines in the correct sequence:

```
Discovery → Platform Core → Validation → Platform Core → Scoring → Platform Core → Governance → Platform Core → Export
```

The engines remain independent. The Platform Core determines execution order.

2. Control Dependency Management

Manages the Directed Acyclic Graph (DAG) of control dependencies:



Dependencies are defined in `config.yaml`:

```
control_dependencies:
  C02: [C01]
  C09: [C03]
```

3. Engine Registry

Maintains metadata for every registered engine:

Engine	Status	Version	Capabilities
Discovery	Healthy	1.0	Schema discovery, metadata extraction
Validation	Healthy	1.0	10 structured controls (C01-C010)
Scoring	Healthy	1.0	Pass/fail, aggregate scoring
Governance	Healthy	1.0	Release gates, audit, export

4. State Management

The Platform Core owns workflow state:

```
Pipeline Status: Running
├─ Discovery: Complete (passed 100%)
├─ Validation: Running (5/10 controls complete)
├─ Scoring: Pending
└─ Governance: Pending
```

This provides restart capability, checkpointing, resilience, and recovery.

5. Configuration Service

Central configuration management via `config.yaml`:

```
execution:
  environment: DEV
  client_name: DemoBank
  control_id: null          # null = run all controls

release_gate:
  enabled: true
  block_on_status: ["FAIL", "BLOCKED", "ERROR"]
  minimum_score: 80
  enforcement_mode: "STRICT"

rules:
  C01: enabled
  C02: enabled
  # ... through to C010
```

Every engine reads configuration from one location — the Platform Core.

6. Security Enforcement

The Platform Core validates every request:

- Authentication (API keys, environment-based)
- Authorisation (role-based execution permissions)
- Audit logging (every action logged with batch ID and timestamp)
- Secrets isolation (credentials via environment variables, not code)
- Configuration validation (pre-execution checks)

7. Error Handling & Failure Isolation

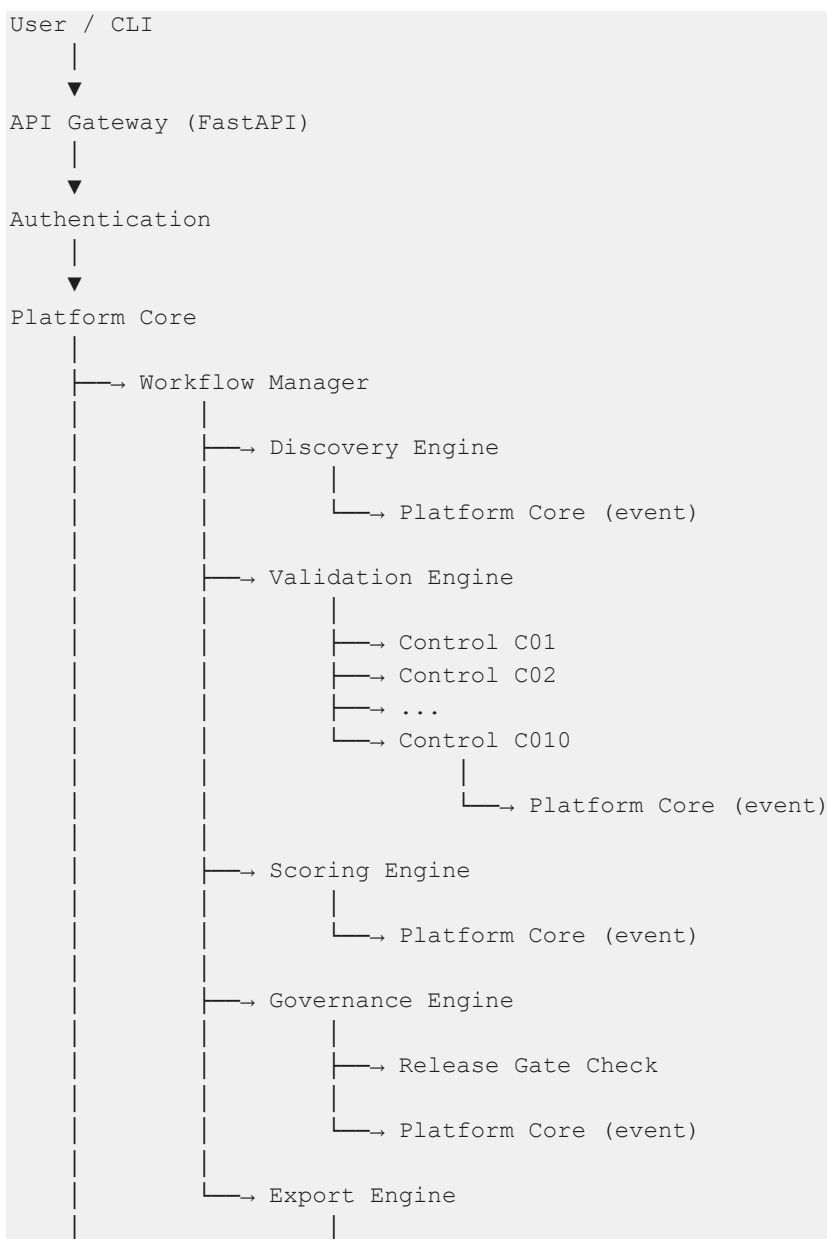
- **Control-level isolation** — A failure in C01 does not block C03
- **Graceful degradation** — Non-critical failures logged, execution continues
- **Retry logic** — Configurable timeout per control (default: 300 seconds)
- **Exception register** — All failures recorded for audit
- **Recovery mode** — Re-run only failed controls from previous batch

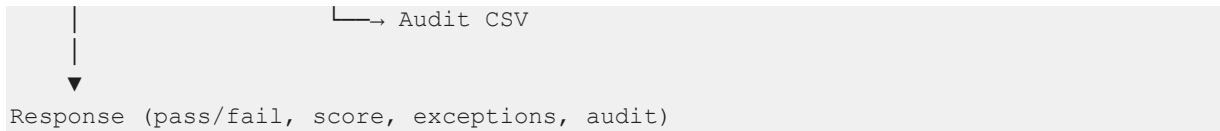
8. Observability

Provides a single operational view:

Metric	Source
Engine execution time	Platform Core timing
Control pass/fail rates	Scoring engine results
Batch duration	Workflow start/end timestamps
Exception counts	Exception register
API latency	FastAPI middleware

Interaction Model





No engine communicates directly with another. All coordination passes through the Platform Core.

Current Implementation

The Platform Core is implemented in `app/execution_engine.py`:

```

class ExecutionEngine:
    def __init__(self, config, batch_id=None):
        self.config = config
        self.batch_id = batch_id or str(uuid.uuid4())
        self.recovery_mode = False

    def run(self):
        # 1. Initialise batch in database
        # 2. Resolve control execution order (DAG)
        # 3. Execute each control in dependency order
        # 4. Score results
        # 5. Apply release gates
        # 6. Export audit trail
  
```

Key files:

- `app/execution_engine.py` — Core orchestrator
- `app/rule_executor.py` — Individual control execution
- `app/rule_factory.py` — Control instantiation from config
- `app/scoring_engine.py` — Scoring logic
- `app/audit_export.py` — CSV audit export
- `config.yaml` — Centralised configuration
- `app/db_connector.py` — Database connectivity

Azure Alignment

The Platform Core maps naturally to Azure services:

Platform Core Function	Azure Service
Compute / Hosting	Azure Container Apps
API Gateway	Azure API Management
Identity	Microsoft Entra ID

Secrets	Azure Key Vault
Monitoring	Azure Monitor / Application Insights
Database	Azure SQL Database / PostgreSQL Flexible Server
Storage	Azure Blob Storage
CI/CD	GitHub Actions / Azure DevOps
Infrastructure	Bicep / ARM Templates

Architectural Principles

13. **Loose coupling** — Engines communicate only through the Platform Core
 14. **High cohesion** — Each engine has a single, well-defined responsibility
 15. **Deterministic execution** — Same inputs produce same outputs
 16. **Stateless engines** — State managed by Platform Core
 17. **Centralised governance** — Release gates, audit, and compliance enforced centrally
 18. **Cloud-native** — Designed for Azure, deployable anywhere
 19. **Extensibility** — New engines can be added without modifying existing ones
 20. **Security-by-design** — Authentication, authorisation, and audit built into the core
 21. **Failure isolation** — One control failure does not cascade
 22. **Observability-by-default** — Every execution produces audit evidence
-

Benefits

Technical

- Reduced coupling between components
- Easier maintenance and testing
- Independent engine evolution
- Simplified debugging (single coordination point)
- Improved resilience and recovery

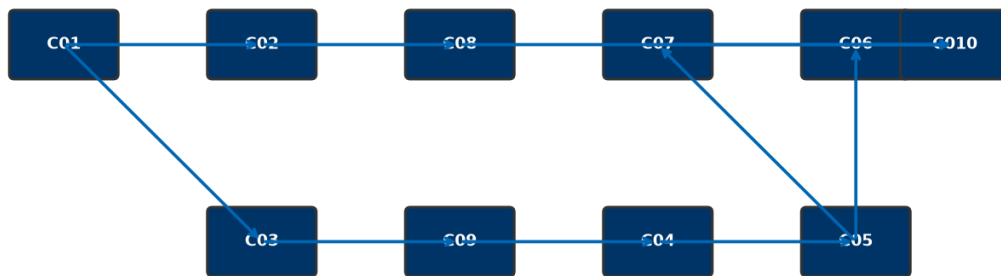
Operational

- Centralised monitoring and alerting
- Consistent configuration management
- Unified security enforcement
- Simplified deployment model
- Faster troubleshooting

Business

- Faster feature delivery (add engines without redesign)
- Lower maintenance costs
- Easier partner integration
- Enterprise-grade governance built-in
- Clear Azure alignment for cloud migration

Control Dependency DAG (C01-C010)



Control Dag

Phase 1.4 — Azure Cloud Architecture

Version: 2.0

Product: FS Migration Validation Engine

Status: Founders Hub Ready

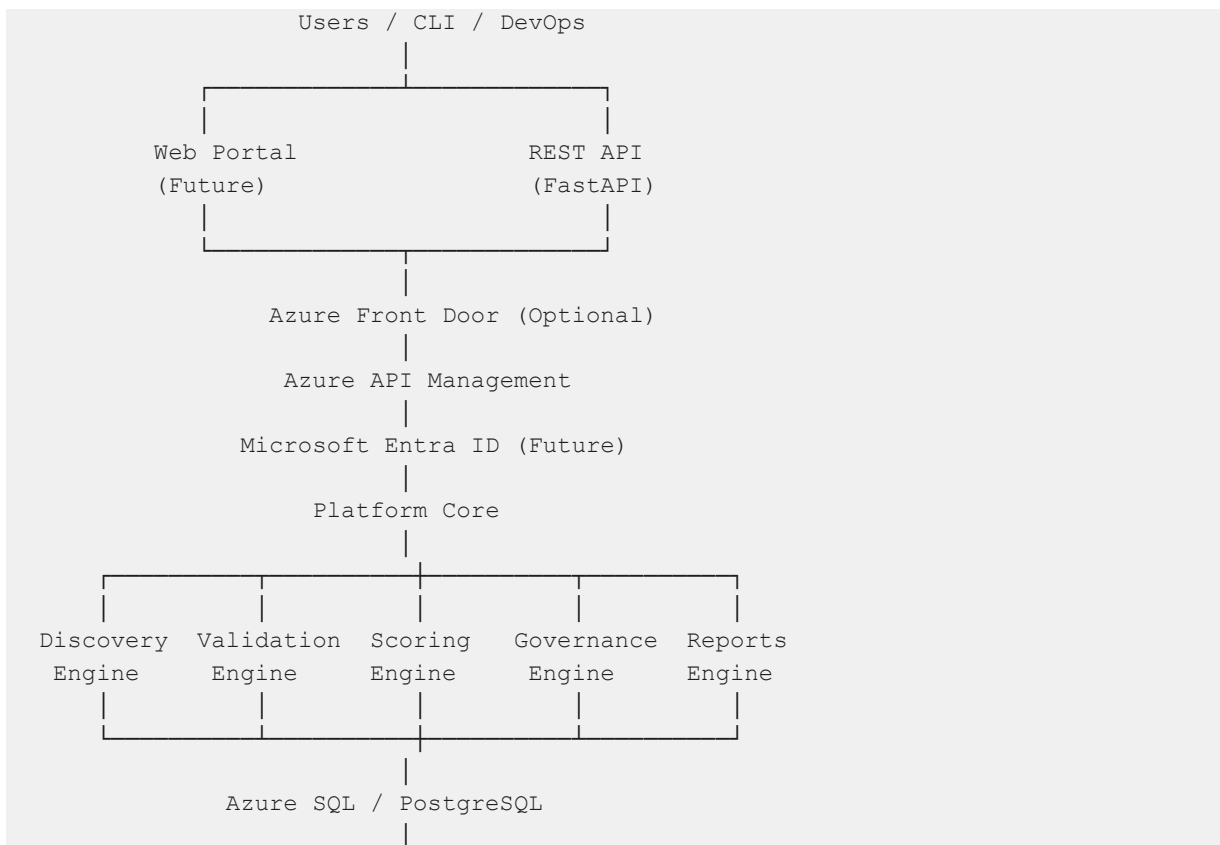
Audience: Microsoft Reviewers, Cloud Architects, Engineering Team

Executive Summary

The FS Migration Validation Engine is designed as an **Azure-native, cloud-first SaaS solution** built on Microsoft's Well-Architected Framework. It leverages managed Azure services to maximise scalability, security, reliability, and operational efficiency for regulated Financial Services data migration validation.

The architecture follows a **Platform Core + microservices** model where independent domain engines (Discovery, Validation, Scoring, Governance) are orchestrated through a central coordination layer. The solution is cloud-portable while taking full advantage of Azure's managed services ecosystem.

High-Level Architecture



Core Azure Services

Azure Container Apps (Recommended MVP)

The primary compute platform for the validation engine.

Advantages:

- Serverless container hosting
- Automatic scaling based on validation workload
- Minimal operational overhead
- Cost-efficient consumption pricing
- Native Dapr support for future microservices
- Fast deployment from Docker images

Migration path: Azure Kubernetes Service (AKS) for enterprise-scale deployments.

Azure API Management

Secure gateway for all API interactions.

Responsibilities:

- API gateway for REST endpoints
- Rate limiting and throttling
- API versioning
- Authentication enforcement (future: OAuth2 / Entra ID)
- Request validation
- Usage analytics
- Developer portal for integration documentation

Microsoft Entra ID (Future Integration)

Identity and access management for enterprise deployments.

Capabilities:

- Single Sign-On (SSO)
- OAuth2 / OpenID Connect
- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)
- Conditional Access policies
- Enterprise identity federation

Azure SQL Database / PostgreSQL Flexible Server

Primary data store for the validation engine.

Stores:

- Migration batch records
- Control execution results
- Exception register
- Audit trail
- Configuration metadata
- Customer and project data (future)

Benefits:

- Managed service with automated backups
- High availability with zone redundancy
- Built-in threat detection (Defender for Cloud)
- Encryption at rest and in transit

Azure Blob Storage

Secure storage for audit artifacts and exports.

Stores:

- Audit CSV exports
- Validation reports
- Uploaded schema definitions
- Log archives
- Backup snapshots

Benefits:

- Cost-effective tiered storage
- Lifecycle management policies
- Immutable storage for compliance
- Geo-redundant replication

Azure Key Vault

Centralised secrets management.

Stores:

- Database connection strings
- API keys
- Certificates
- Encryption keys
- Environment credentials

Security:

- No secrets in source code or configuration files
- Managed identities for service-to-service authentication
- Access policies with least privilege
- Audit logging for all secret access

Azure Monitor & Application Insights

Comprehensive observability.

Captures:

- Infrastructure metrics (CPU, memory, network)
- API response times and error rates
- Control execution duration
- Batch processing performance
- Exception tracking
- Distributed tracing across microservices
- Custom dashboards and alerts

GitHub Actions / Azure DevOps

CI/CD pipeline automation.

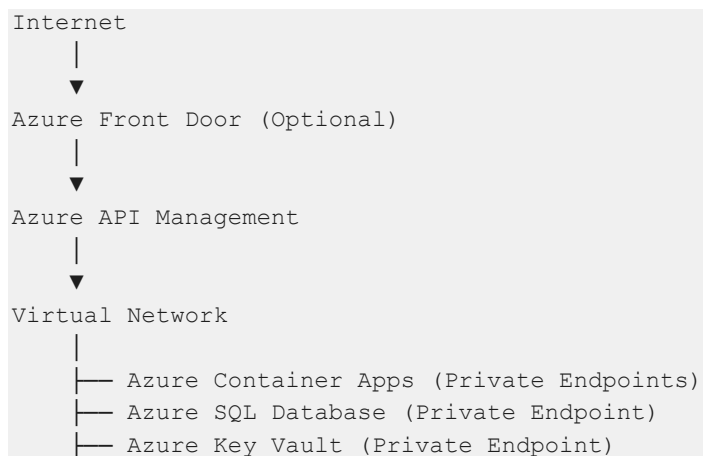
Pipeline stages:

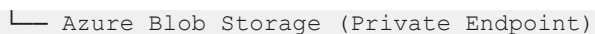
Commit → Build → Unit Tests → Security Scan → Container Build → Deploy to Test → Integration Tests → Deploy to Production

Infrastructure as Code:

- Bicep / ARM Templates for Azure resource provisioning
- Version-controlled infrastructure definitions
- Reproducible deployments across environments

Networking Architecture



 Azure Blob Storage (Private Endpoint)**Security features:**

- Private networking for all data services
- Network Security Groups (NSGs)
- DDoS protection
- TLS 1.2+ for all communications
- Managed certificates via Azure Key Vault
- Service endpoints / Private Link for PaaS services

Security Architecture

The platform follows a **Zero Trust** security model:

Security Layer	Implementation
Identity	Microsoft Entra ID + RBAC (future)
Authentication	API keys → OAuth2 / Entra ID (migration path)
Secrets	Azure Key Vault, no secrets in code
Network	Private endpoints, TLS, NSGs
Data at rest	Azure SQL TDE, Blob encryption
Data in transit	TLS 1.2+ for all connections
Monitoring	Defender for Cloud, Sentinel (enterprise)
Auditing	Full audit trail for every execution

Deployment Environments

Environment	Compute	Database	Purpose
Development	Local Docker	Local PostgreSQL	Local development and testing
Test/QA	Azure Container Apps	Azure SQL (shared)	Automated testing, CI/CD
Staging	Azure Container Apps	Azure SQL (isolated)	Pre-production validation
Production	Azure Container Apps	Azure SQL (HA)	Customer deployments

	(MVP) → AKS (scale)		
--	---------------------	--	--

Scalability Strategy

Each engine scales independently based on workload:

Engine	Scale Trigger
API Layer	Concurrent API requests
Validation Engine	Number of controls × batch size
Scoring Engine	Post-validation scoring requests
Audit Export	Export request volume

Autoscaling is managed by Azure Container Apps using:

- CPU utilisation thresholds
- Memory utilisation thresholds
- HTTP request queue depth
- Custom scaling rules (future)

High Availability

Azure services provide enterprise-grade availability:

- Zone-redundant deployment (Azure Container Apps)
- Active geo-replication (Azure SQL Database)
- Automatic failover with managed services
- Automated backups with point-in-time restore
- Health probes for container instances
- Rolling updates for zero-downtime deployments

Azure Well-Architected Framework Alignment

Reliability

- Managed Azure services with built-in redundancy
- Automated health monitoring and recovery

- Fault-tolerant architecture (failure isolation per control)
- Automated backups and point-in-time restore

Security

- Zero Trust architecture
- Microsoft Entra ID + Azure Key Vault
- Private endpoints for data services
- Defender for Cloud continuous assessment
- Full audit trail for compliance

Cost Optimisation

- Consumption-based pricing (Azure Container Apps)
- Autoscaling to match workload demand
- Managed PaaS services reduce operational overhead
- Blob storage lifecycle policies for cost-effective archiving
- Right-sizing through Azure Advisor recommendations

Operational Excellence

- CI/CD with GitHub Actions / Azure DevOps
- Infrastructure as Code (Bicep / ARM)
- Centralised monitoring (Azure Monitor)
- Automated deployments with quality gates
- Comprehensive audit logging

Performance Efficiency

- Stateless API design for horizontal scaling
- Independent engine scaling
- Connection pooling for database efficiency
- Background processing for long-running validations
- Caching strategies for metadata (future)

Azure Service Mapping

Platform Capability	Azure Service
Compute (MVP)	Azure Container Apps
Compute (Enterprise)	Azure Kubernetes Service (AKS)
API Gateway	Azure API Management

Global Routing	Azure Front Door
Identity	Microsoft Entra ID
Primary Database	Azure SQL Database / PostgreSQL Flexible Server
Secrets Management	Azure Key Vault
Object Storage	Azure Blob Storage
Monitoring	Azure Monitor
Application Telemetry	Application Insights
Security Posture	Microsoft Defender for Cloud
CI/CD	GitHub Actions / Azure DevOps
Infrastructure as Code	Bicep / ARM Templates

Future Azure Services

Phase	Service	Purpose
Phase 2	Azure OpenAI Service	AI-assisted mapping recommendations
Phase 2	Azure Service Bus	Async event processing for large batches
Phase 3	Azure Cosmos DB	Global-scale metadata storage
Phase 3	Azure Cognitive Search	Semantic search across mappings
Phase 3	Power BI	Interactive dashboards and reporting
Phase 4	Microsoft Fabric	Enterprise data integration

Cost Model (Projected)

Service	Estimated Monthly Cost (MVP)
Azure Container Apps	\$200–500

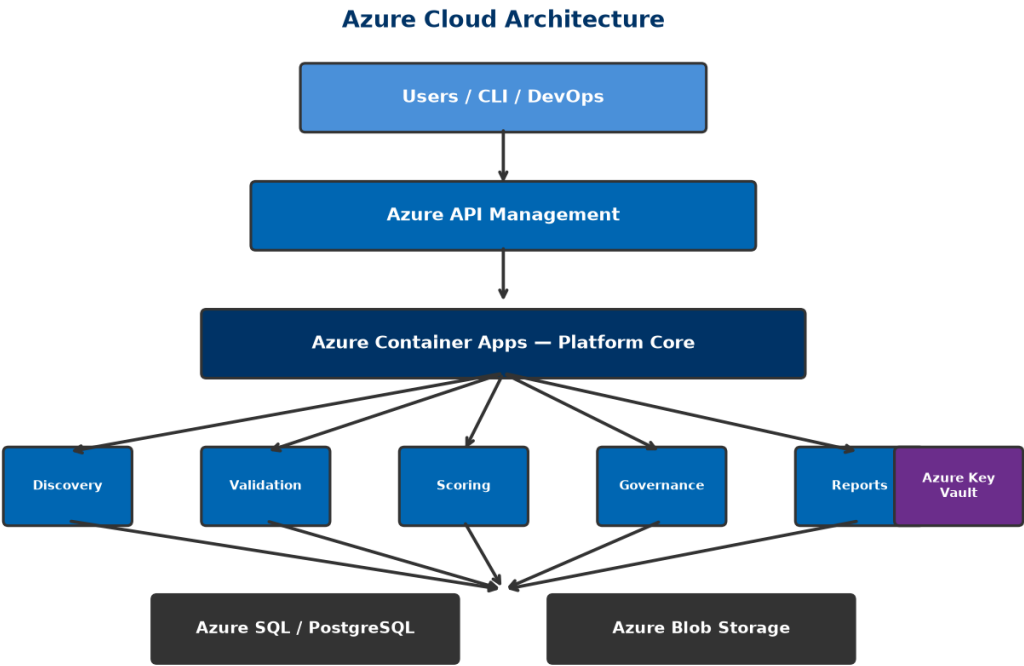
Azure SQL Database (Basic)	\$150–300
Azure Blob Storage	\$50–100
Azure API Management	\$100–200
Azure Key Vault	\$10–20
Azure Monitor	\$50–100
Total (MVP)	\$560–1,220/month

Costs scale linearly with customer adoption. Enterprise deployments with AKS and HA configurations range from \$2,000–5,000/month.

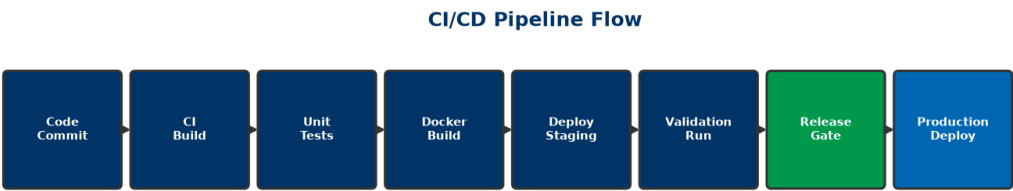
Conclusion

This Azure-native architecture provides a secure, scalable, and cost-effective foundation for the FS Migration Validation Engine. By combining a centralised Platform Core with independently deployable microservices and managed Azure services, the platform minimises operational overhead while maximising resilience, extensibility, and enterprise readiness for regulated Financial Services environments.

The design supports rapid MVP deployment through Azure Container Apps while providing a clear migration path to Azure Kubernetes Service for large-scale enterprise deployments. It establishes a robust foundation for future AI capabilities, multi-tenant SaaS operations, and a growing ecosystem of platform integrations.



Azure Architecture



Cicd Pipeline

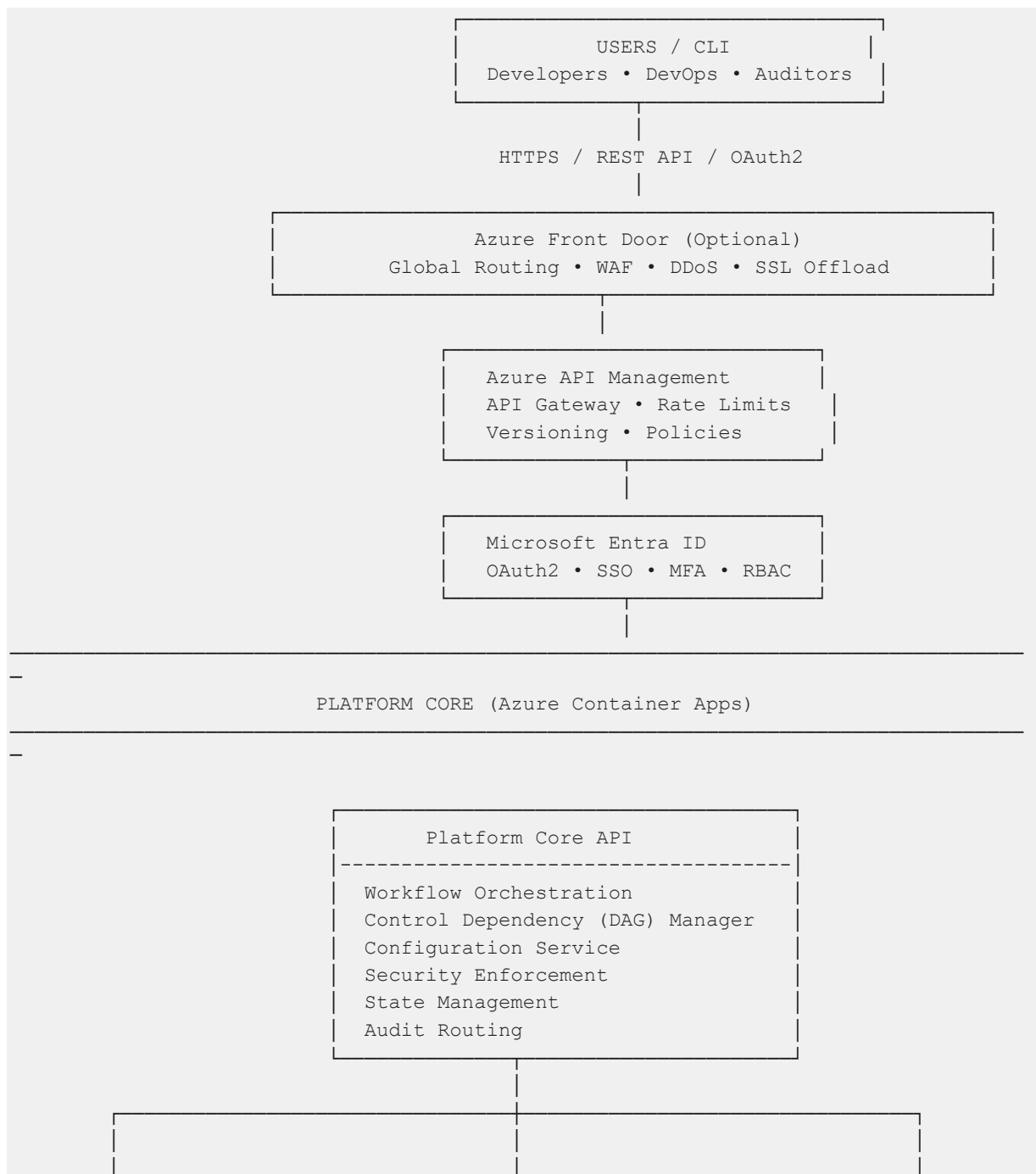
Phase 1.5 — Azure Reference Architecture Diagram

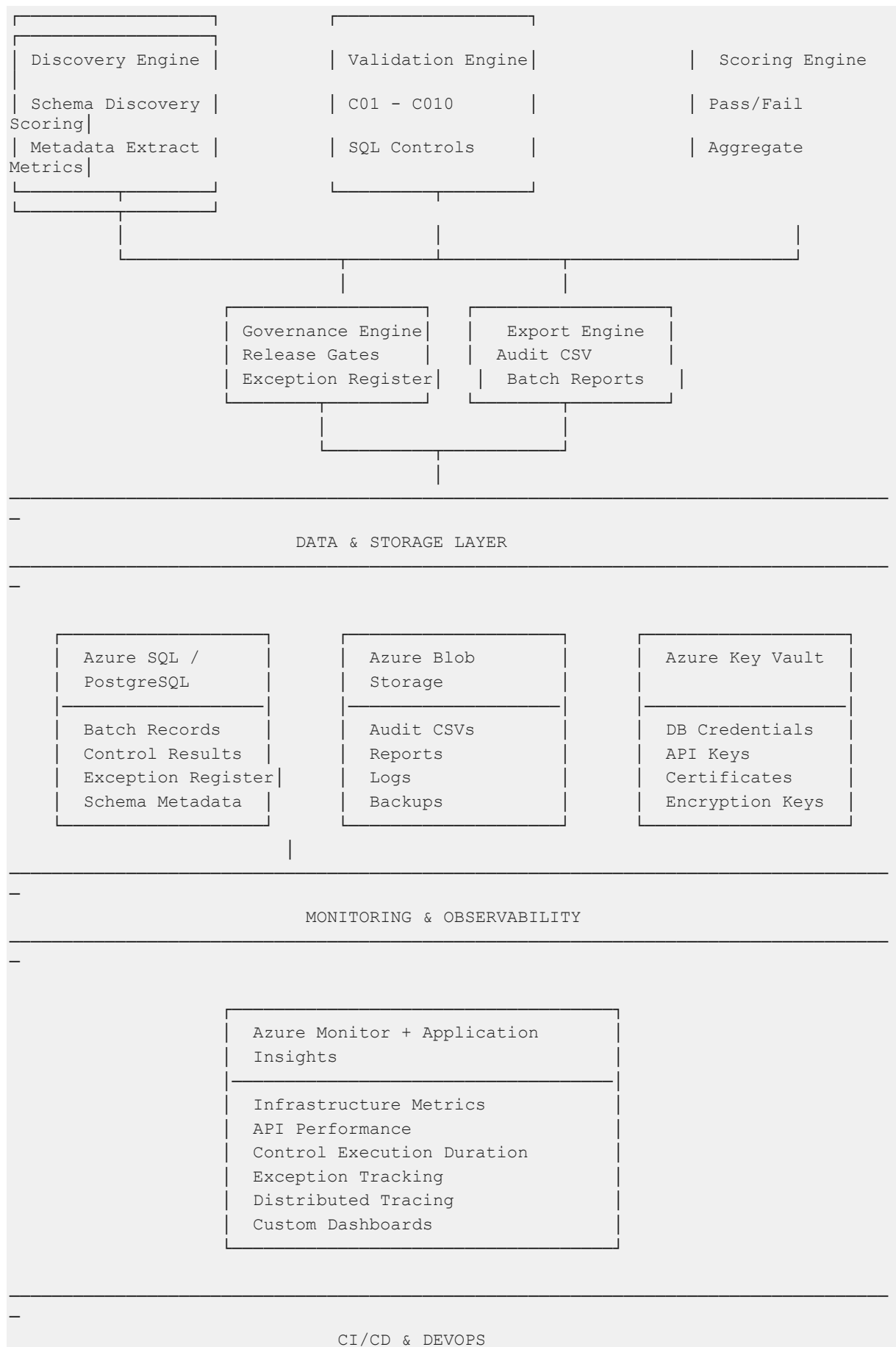
Version: 2.0

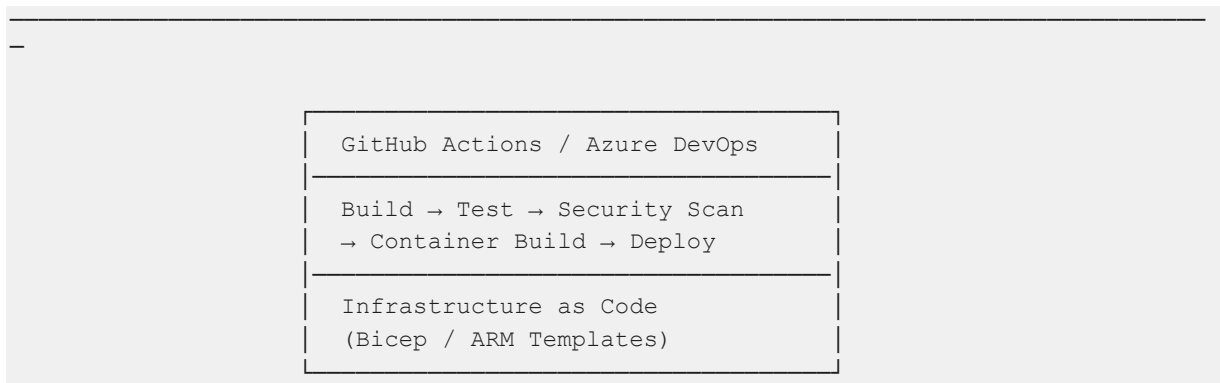
Architecture Style: Azure-Native • Microservices • Event-Driven • Financial Services Grade

Product: FS Migration Validation Engine

Azure Reference Architecture







Architectural Layers

1. Presentation Layer

- **CLI Interface** — `python app/main.py run --config config.yaml`
- **REST API** — FastAPI endpoints for programmatic access
- **Future Web Portal** — Interactive dashboard and management UI

2. API Layer

- Azure API Management — Gateway, rate limiting, versioning
- Microsoft Entra ID — Authentication and authorisation (future)
- Azure Front Door — Global routing and WAF (optional)

3. Platform Core

- Workflow orchestration — Coordinates engine execution sequence
- Control dependency management — DAG-based execution ordering
- Configuration management — Centralised via `config.yaml`
- Security enforcement — Authentication, audit, secrets isolation
- State management — Batch tracking, checkpointing, recovery
- Error handling — Failure isolation, retry, graceful degradation

4. Domain Engines

Each engine is independently deployable and scalable:

Engine	Function	Status
Discovery Engine	Schema discovery, metadata extraction	[OK] Built
Validation Engine	10 structured controls (C01-C010)	[OK] Built

Scoring Engine	Pass/fail, aggregate scoring	[OK] Built
Governance Engine	Release gates, exception register	[OK] Built
Export Engine	CSV audit export	[OK] Built

5. Data Layer

- **Azure SQL / PostgreSQL** — Primary operational database
- **Azure Blob Storage** — Audit artifacts and report storage
- **Azure Key Vault** — Secrets and credentials management

6. Observability Layer

- Azure Monitor — Infrastructure and platform metrics
- Application Insights — Application performance monitoring
- Custom dashboards — Migration quality visualisation

7. DevOps Layer

- GitHub Actions / Azure DevOps — CI/CD pipeline automation
- Bicep / ARM Templates — Infrastructure as Code
- Automated testing — Unit tests, integration tests, security scans

Deployment Model

Environment	Compute	Database	Storage	Configuration
Development	Local Docker	Local PostgreSQL	Local filesystem	`config.yaml` + `.env`
Test	Azure Container Apps	Azure SQL (DTU)	Blob Storage (Cool)	Key Vault + Config
Staging	Azure Container Apps	Azure SQL (vCore)	Blob Storage (Cool)	Key Vault + Config
Production	AKS (future)	Azure SQL (HA)	Blob Storage (Hot)	Key Vault + Config

Security Boundaries

Boundary	Control
Network	Private endpoints for all data services
Identity	Environment-based → Microsoft Entra ID (future)
Authentication	API keys → OAuth2 (migration path)
Secrets	Azure Key Vault — no secrets in code
Data at rest	Azure SQL TDE, Blob encryption
Data in transit	TLS 1.2+ for all communications
Audit	Every execution logged with batch ID and timestamp

Scalability Model

Component	Scale Strategy	Metric
API Layer	Horizontal (Container Apps)	Requests/second
Validation Engine	Horizontal	Controls × batch size
Scoring Engine	Vertical	Result volume
governance Engine	Singleton	Batch completion events
Database	Vertical → Read replicas	Connection pool depth

Future Expansion

The architecture is designed to support:

- **Phase 2:** AI-assisted mapping (Azure OpenAI), async processing (Service Bus)
- **Phase 3:** Web dashboard (React + Azure Static Web Apps), multi-tenant SaaS
- **Phase 4:** Marketplace, partner integrations, Microsoft Fabric connectivity

Phase 1.6 — Azure Founders Hub Technical Narrative

Version: 2.0

Program: Microsoft for Startups Founders Hub

Product: FS Migration Validation Engine

Executive Summary

Our platform is an **automated, metadata-driven data migration validation and governance platform** purpose-built for regulated Financial Services institutions. It systematically validates data migrations across core banking, payments, lending, asset management, and regulatory reporting systems — ensuring accuracy, completeness, and regulatory compliance.

The solution is designed as an Azure-native, cloud-first platform that combines Microsoft's managed cloud services with intelligent automation to reduce the cost, complexity, and risk of financial data migrations. By adopting managed Azure services, a modular Platform Core architecture, and metadata-driven validation, we enable financial institutions to validate migrations with complete coverage, objective scoring, and auditable governance.

Business Problem

Financial Services institutions undertaking data migration projects face critical challenges:

Challenge	Impact
Regulatory scrutiny	FCA, PRA, Basel, and SOX require demonstrable data integrity
Manual validation	60%+ of validation effort is manual, subjective, and error-prone
Incomplete coverage	Sampling-based checks miss edge cases and exceptions
No repeatability	Every migration project reinvents validation from scratch
Poor audit trails	Spreadsheet tracking fails regulatory due diligence
High failure cost	Data errors cause operational losses, regulatory fines, reputational damage

These challenges are particularly acute in Financial Services, where a single data migration error can trigger regulatory penalties, operational disruptions, and significant financial loss.

Our Solution

The FS Migration Validation Engine provides:

23. **Automated Schema Discovery** — Connect to source and target databases, discover schemas, extract metadata
24. **10 Structured Validation Controls** — Comprehensive validation framework covering row counts, data types, nullability, keys, referential integrity, business rules, date boundaries, numeric precision, string patterns, and regulatory completeness
25. **Objective Quality Scoring** — Automated pass/fail scoring with configurable thresholds
26. **Release Gate Governance** — Block migrations that fail to meet minimum quality standards
27. **Audit-Ready Exports** — CSV audit trails suitable for regulatory review
28. **REST API** — Programmatic integration with CI/CD pipelines and deployment workflows

Why Azure

Microsoft Azure provides the enterprise-grade foundation required for regulated Financial Services deployments:

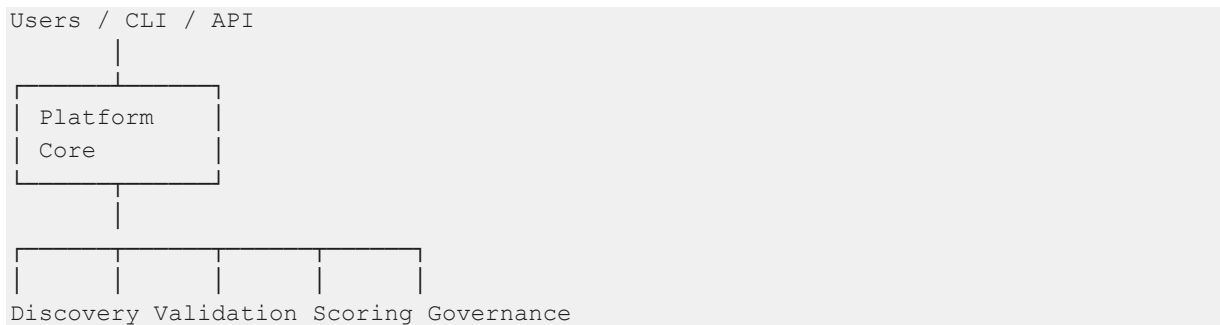
Requirement	Azure Service
Scalable compute	Azure Container Apps — serverless, autoscaling
Managed database	Azure SQL Database — high availability, automated backups
Secure storage	Azure Blob Storage — encrypted, tiered, immutable storage
Secrets management	Azure Key Vault — no credentials in code
Identity & access	Microsoft Entra ID — enterprise-grade IAM
API management	Azure API Management — gateway, throttling, versioning
Monitoring	Azure Monitor + Application Insights — observability
Security	Microsoft Defender for Cloud — continuous assessment

Architectural alignment:

- [OK] Azure Well-Architected Framework (all 5 pillars)
- [OK] Cloud Adoption Framework

- [OK] Zero Trust security model
- [OK] Infrastructure as Code (Bicep / ARM)
- [OK] CI/CD with GitHub Actions / Azure DevOps

Platform Architecture



The **Platform Core** orchestrates execution across independent domain engines. Each engine communicates only with the Platform Core, ensuring loose coupling, independent scalability, and centralised governance.

Current implementation status:

Component	Status	Location in Codebase
CLI Engine	[OK] Production-ready	`app/main.py`, `app/execution_engine.py`
10 Validation Controls	[OK] Implemented	`sql/controls/`
Scoring Engine	[OK] Implemented	`app/scoring_engine.py`
Release Gates	[OK] Implemented	`config.yaml` — minimum score enforcement
Audit Export	[OK] Implemented	`app/audit_export.py` — CSV generation
REST API	[OK] Built	`app/api/` — FastAPI endpoints
Docker Deployment	[OK] Ready	`Dockerfile`, `docker-compose.yml`
Database Schema	[OK] Production-ready	`sql/schema/`

Security Strategy

Implemented according to Zero Trust principles:

- **No secrets in code** — All credentials via environment variables → Azure Key Vault
- **Encryption at rest and in transit** — TLS 1.2+, Azure SQL TDE, Blob encryption
- **Audit logging** — Every validation execution logged with batch ID and timestamp
- **Failure isolation** — Control-level isolation prevents cascade failures
- **Release gates** — Automated quality gates block non-compliant migrations

Competitive Advantages

Area	Our Advantage
Purpose-built	Designed for regulated Financial Services, not generic ETL
10 structured controls	Comprehensive, standardised validation framework
Governance-first	Release gates, audit trails, compliance built-in
Objective scoring	Automated, repeatable quality metrics
Docker-native	Deploy anywhere — cloud, on-premises, hybrid
API-first	Integrate with existing CI/CD pipelines
Open architecture	Python-based, extensible, plugin-ready

Scale & Performance

The platform is designed for enterprise-scale migration validation:

- **Database discovery** — Connect to multiple source and target databases
- **Batch processing** — Validate migrations in structured batch workflows
- **Horizontal scaling** — Stateless API design, containerised deployment
- **Control isolation** — Independent execution per control with failure isolation
- **Configurable timeout** — Per-control timeout (default: 300 seconds)

Roadmap

Phase	Capabilities	Timeline
Phase 1	CLI engine, 10 controls, scoring, audit, release gates	[OK] Complete
Phase 2	REST API, dashboard, AI-assisted mapping (Azure OpenAI)	 In Development
Phase 3	Multi-tenant SaaS, web portal, interactive dashboards	 Planned
Phase 4	Partner integrations, Microsoft Fabric, regulatory template library	 Future

Value to Microsoft

This platform demonstrates effective use of the Microsoft ecosystem by:

- Building on **Azure-native managed services** (Container Apps, SQL, Blob, Key Vault)
- Following the **Azure Well-Architected Framework** across all 5 pillars
- Leveraging **Microsoft Entra ID** for enterprise identity (future)
- Supporting **GitHub Actions / Azure DevOps** for CI/CD
- Providing a scalable SaaS foundation suitable for **Azure Marketplace**
- Creating **Azure consumption growth** as customer adoption scales

As customer adoption grows, usage expands across Azure compute, storage, identity, AI, monitoring, and integration services — reinforcing long-term engagement with the Microsoft cloud platform.

Conclusion

The FS Migration Validation Engine represents a modern approach to data migration assurance for regulated Financial Services. Its automated validation framework, metadata-driven architecture, and Azure-native design provide financial institutions with the confidence, governance, and auditability required for mission-critical data migrations.

By aligning with Microsoft's architectural guidance and leveraging Azure's managed services, the platform is positioned to deliver measurable value to enterprise customers while demonstrating strong technical alignment with the goals of the Azure Founders Hub program.

Phase 2.1 — Executive Pitch Narrative

Version: 2.1

Product: FS Migration Validation Engine

Target: Microsoft Founders Hub, Investors, Enterprise Customers

Currency: GBP (£) — UK-based company

Vision

Financial Services institutions migrating mission-critical data between systems face a fundamental challenge: how to ensure accuracy, completeness, and regulatory compliance when millions of records are in motion.

Our vision is to make data migration validation **automated, objective, and audit-ready** — transforming it from a manual, high-risk exercise into a repeatable, governed, and verifiable process.

The Problem

Data migration in Financial Services is broken:

- **Manual validation** — Teams spend weeks manually sampling and checking data
- **Incomplete coverage** — Sampling misses edge cases and exceptions
- **No standardisation** — Every migration project reinvents validation from scratch
- **Poor auditability** — Spreadsheets and emails fail regulatory scrutiny
- **High cost of failure** — A single data error can cause operational losses, regulatory fines, and reputational damage

Banks migrating core banking systems, payment platforms, or regulatory reporting systems cannot afford data quality failures. Yet most validation today relies on custom scripts, manual checks, and hope.

Our Solution

The FS Migration Validation Engine is an **automated, metadata-driven validation and governance platform** that:

29. **Discovers** legacy and target system schemas automatically
30. **Validates** data across 10 structured migration controls
31. **Scores** migration quality with objective, repeatable metrics
32. **Governs** the migration lifecycle with release gates and audit trails
33. **Reports** comprehensive audit evidence suitable for regulatory review

How It Works

Legacy System → Discovery → 10 Validation Controls → Scoring → Release Gate → Audit Report

The platform runs as a CLI engine, REST API, or Docker container — integrating into existing CI/CD pipelines and deployment workflows.

Why Now?

Several trends make this the right time for an automated migration validation platform:

Trend	Impact
Regulatory pressure	FCA, PRA, Basel IV, SOX demand demonstrable data integrity
Cloud migration acceleration	Banks are moving core systems to cloud at unprecedented pace
AI expectations	Institutions expect intelligent automation, not manual processes
Legacy system retirement	COBOL, mainframe, and legacy platform expertise is retiring
Cost pressure	Financial institutions need to reduce project costs and timelines

Why Azure?

Microsoft Azure provides the enterprise-grade foundation required for regulated Financial Services:

- **Azure Container Apps** — Scalable, serverless compute
- **Azure SQL Database** — Managed, highly available database
- **Azure Key Vault** — Secure secrets management
- **Microsoft Entra ID** — Enterprise identity and access
- **Azure Monitor** — Comprehensive observability
- **Azure OpenAI** — Future AI-assisted mapping capabilities

Competitive Advantage

Area	Our Advantage
------	---------------

Purpose-built	Designed for regulated Financial Services, not generic ETL
10 structured controls	Comprehensive, standardised validation framework
Governance-first	Release gates, audit trails, compliance built-in
Objective scoring	Automated, repeatable quality metrics
Docker-native	Deploy anywhere — cloud, on-premises, hybrid
API-first	Integrate with existing CI/CD pipelines

Business Model

The platform will be offered as a **SaaS subscription** with tiered pricing:

Tier	Target	Price Range (GBP/month)
Starter	Small migrations, pilot projects	£1,500–£3,500
Professional	Mid-size bank migrations	£3,500–£10,000
Enterprise	Large-scale, multi-system migrations	£10,000–£35,000
Strategic	Enterprise-wide deployment, custom controls	Custom pricing

Typical annual contract value: £30,000–£80,000 per customer

Additional revenue streams:

- Professional services (implementation, custom controls at £5k–£20k each)
- AI-assisted mapping module (future)
- Enterprise support and SLAs

Go-to-Market

Primary channels:

- Microsoft Founders Hub and Azure Marketplace
- Direct enterprise sales to UK Financial Services CTOs and Heads of Data
- System integrator partnerships (Accenture, Deloitte UK, Infosys UK)
- Microsoft partner network and co-sell programme

Target customers:

- Tier 1–3 Banks (Barclays, Lloyds, NatWest, challenger banks)
- Insurance companies (Aviva, Legal & General, AXA UK)
- Asset management firms
- FinTech and payment processors
- Regulated enterprises undergoing data migration

Product Roadmap

Phase	Capabilities	Timeline
Phase 1	CLI engine, 10 controls, scoring, audit, release gates	[OK] Complete
Phase 2	REST API, dashboard, AI-assisted mapping (Azure OpenAI)	 In Development
Phase 3	Multi-tenant SaaS, web portal, interactive dashboards	 Planned
Phase 4	Partner integrations, Microsoft Fabric, regulatory template library	 Future

Investment Opportunity

Azure Founders Hub support will accelerate:

- **Product development** — Complete Phase 2 (API, dashboard, AI mapping)
- **Customer validation** — Secure 5+ pilot customers in UK Financial Services
- **Security certifications** — ISO 27001, SOC 2 readiness
- **Marketplace readiness** — Azure Marketplace listing
- **Commercial growth** — Early customer acquisition targeting £450k ARR

Funding ask for Founders Hub: Azure credits (£120k max) + partner programme support

Seed round (if required): £500,000–£750,000

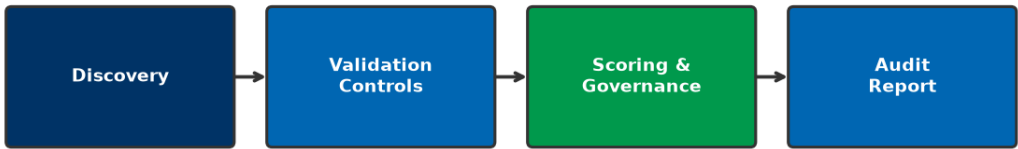
The platform is designed to grow Azure consumption as customers scale their migration initiatives, creating long-term value for both customers and the Microsoft ecosystem.

Closing Statement

Our mission is to transform data migration validation from a slow, manual, high-risk exercise into an automated, governed, and audit-ready process. By combining Azure-native architecture with

intelligent automation, we aim to become the standard platform for regulated data migration assurance — helping Financial Services institutions migrate with confidence, compliance, and speed.

Business Flow



Business Flow

Phase 2.2 — Investor Pitch Deck

Version: 2.1

Product: FS Migration Validation Engine

Recommended Length: 14 Slides

Currency: GBP (£) — UK-based company

Slide 1 — Title

Data Migration Validation for Regulated Financial Services

FS Migration Validation Engine — Automated, Governed, Audit-Ready

Built on Microsoft Azure

"Making data migration validation automated, objective, and compliant."

Slide 2 — The Problem

Data Migration Validation Is Broken

Financial Services institutions rely on:

- [X] Manual sampling (weeks of effort, misses edge cases)
- [X] Spreadsheet tracking (fails regulatory scrutiny)
- [X] Custom scripts (not repeatable, not standardised)
- [X] Gut feel and hope (not objective, not auditable)

Consequences:

- Regulatory fines for data integrity failures
 - Operational losses from undetected errors
 - Migration delays and budget overruns
 - Reputational damage
-

Slide 3 — Market Opportunity

A Growing Regulatory Imperative

Target Market: Financial Services — Banks, Insurers, Asset Managers, FinTech

Market Drivers:

- Regulatory pressure (FCA, PRA, Basel IV, SOX)
- Accelerated cloud migration in Financial Services

- Legacy system retirement (COBOL, mainframe)
- AI and automation expectations

Market Size: Global data migration market projected to grow significantly as Financial Services accelerates cloud adoption.

Slide 4 — Our Solution

Automated Migration Validation Platform

The FS Migration Validation Engine provides:

- [OK] **Automated schema discovery** — Connect, discover, and compare
- [OK] **10 structured validation controls** — Comprehensive, standardised
- [OK] **Objective scoring** — Automated pass/fail with configurable thresholds
- [OK] **Release gate governance** — Block bad migrations before they reach production
- [OK] **Audit-ready exports** — CSV trails suitable for regulatory review

Slide 5 — How It Works

Automated Validation Workflow

```
Legacy System
↓
Schema Discovery (automated)
↓
10 Validation Controls (C01-C010)
↓
Objective Scoring (pass/fail per control)
↓
Release Gate (min 80% score required)
↓
Audit Report (CSV export)
```

The platform runs as CLI, REST API, or Docker — integrating into any CI/CD pipeline.

Slide 6 — Why We're Different

Feature	Manual / Custom	Our Platform
Schema discovery	Manual analysis	[OK] Automated
Validation coverage	Sampling (~20%)	[OK] 100% of records
Scoring	Subjective	[OK] Objective metrics

Governance	Spreadsheet tracking	[OK] Release gates + audit
Repeatability	None (new scripts each time)	[OK] Standardised framework
Regulatory audit	Manual evidence gathering	[OK] Automated CSV export

Slide 7 — Technology

Azure-Native Architecture

Built on Microsoft Azure:

- **Azure Container Apps** — Serverless compute
- **Azure SQL Database** — Managed, HA database
- **Azure Key Vault** — Secrets management
- **Microsoft Entra ID** — Enterprise IAM (future)
- **Azure Monitor** — Observability
- **Azure API Management** — API gateway

Architecture: Platform Core + microservices — modular, extensible, cloud-native.

Slide 8 — Business Model

SaaS Subscription Pricing

Tier	Target Customer	Price (GBP/month)
Starter	Pilot projects, small migrations	£1,500–£3,500
Professional	Mid-size bank migrations	£3,500–£10,000
Enterprise	Large-scale migrations	£10,000–£35,000
Strategic	Enterprise-wide deployment	Custom

Additional: Professional services, AI mapping module (future), support/SLAs.

Slide 9 — Go-to-Market

Channel Strategy

Primary:

- Microsoft Founders Hub → Azure Marketplace

- Direct enterprise sales (CTOs, Heads of Data)
- System integrator partnerships (Accenture, Deloitte, Infosys)

Target Verticals:

- Tier 1–3 Banks
- Insurance companies
- Asset management
- FinTech / payments
- Regulated enterprises

Slide 10 — Product Roadmap

Growth Plan

Phase	Capabilities	Status
Phase 1	CLI engine, 10 controls, scoring, audit	[OK] Complete
Phase 2	REST API, dashboard, AI mapping (Azure OpenAI)	 In Development
Phase 3	Multi-tenant SaaS, web portal	 Planned
Phase 4	Marketplace, Fabric integration, regulatory templates	 Future

Slide 11 — Why Microsoft

Built for Azure

Our platform increases Azure consumption through:

- Azure Container Apps
- Azure SQL Database
- Azure Blob Storage
- Azure Key Vault
- Azure OpenAI (future)
- Azure Monitor

Alignment:

- [OK] Azure Well-Architected Framework
- [OK] Cloud Adoption Framework
- [OK] Zero Trust security

- [OK] Marketplace readiness
-

Slide 12 — Investment Opportunity

Use of Funds

Support from Founders Hub will accelerate:

Area	Allocation
Product development (Phase 2)	40%
AI mapping module (Azure OpenAI)	20%
Customer pilots & validation	20%
Security certifications	10%
Marketplace readiness	10%

Slide 13 — Vision

The Future

Our goal is to become the **standard platform for regulated data migration assurance**.

Future capabilities:

- AI-assisted mapping recommendations
 - Intelligent mapping engine with confidence scoring
 - Interactive dashboards
 - Regulatory template library (Basel, SOX, FCA, PRA)
 - Multi-cloud support (AWS, GCP)
-

Slide 14 — Closing

Migration Validation Starts with Certainty

"Don't guess your migration quality — validate it."

FS Migration Validation Engine

Built on Microsoft Azure

Contact: [Your Contact Information]

Phase 2.3 — Market Analysis

TAM • SAM • SOM • Competitor Landscape

Version: 2.0

Product: FS Migration Validation Engine

Target: Financial Services Data Migration Validation Market

Executive Summary

The FS Migration Validation Engine addresses a growing and critical market: **automated data migration validation for regulated Financial Services**. As banks and financial institutions accelerate cloud migration, the need for automated, governed, and audit-ready validation becomes a regulatory and operational necessity.

The market is distinct from generic ETL, data quality, or testing tools — it sits at the intersection of **RegTech (Regulatory Technology)**, **data migration**, and **enterprise governance**.

Market Drivers

Regulatory Pressure

Financial regulators globally are demanding demonstrable data integrity:

- **FCA (UK)** — Operational resilience and data integrity requirements
- **PRA (UK)** — Prudential regulation data standards
- **Basel IV** — Enhanced data quality and reporting requirements
- **SOX (US)** — Financial reporting data integrity
- **GDPR** — Data protection during migration

Cloud Migration Acceleration

Financial Services cloud adoption is accelerating:

- 70%+ of banks have active cloud migration programmes
- Core banking system modernisation is a top 3 priority
- Regulatory reporting systems moving to cloud-native platforms

Legacy System Retirement

Critical legacy platforms are losing expert support:

- COBOL, mainframe, PowerBuilder, Oracle Forms

- Retiring workforce with platform-specific knowledge
- Urgent need for automated migration assurance

AI & Automation Expectations

Financial institutions expect intelligent automation:

- Manual validation is no longer acceptable
- AI-assisted mapping and validation is the expected standard
- Governance and audit must be automated, not manual

Target Market

Primary Customers

Segment	Examples	Pain Point
Tier 1 Banks	HSBC, Barclays, Lloyds, NatWest	Multi-system migrations, regulatory scrutiny
Tier 2–3 Banks	Metro Bank, Virgin Money, challenger banks	Migration cost and risk reduction
Insurance	Aviva, Legal & General, AXA	Policy and claims data migration
Asset Management	BlackRock, Schroders, Legal & General IM	Fund data accuracy during migration
FinTech / Payments	Wise, Revolut, Checkout.com	Rapid migration cycles, compliance requirements
Regulated Enterprises	Healthcare, Government, Utilities	Data integrity obligations

Secondary Customers

Segment	Opportunity
System Integrators	Accenture, Deloitte, Infosys — embed as migration assurance tool
Cloud Migration Practices	Microsoft consulting partners, cloud migration specialists
Audit & Compliance Firms	Big 4 audit firms — use for migration audit evidence

Total Addressable Market (TAM)

The TAM encompasses global spend on:

- Financial Services data migration projects
- RegTech solutions for data governance and compliance
- Enterprise data quality and validation tools
- Cloud migration assurance services

Estimated global TAM: \$8–12 billion annually

Note: Independent analyst data (Gartner, IDC) should be sourced for final figures.

Serviceable Available Market (SAM)

The SAM focuses on organisations that:

- Operate within the Microsoft Azure ecosystem
- Are undertaking active data migration projects
- Have regulatory compliance requirements
- Prefer SaaS-based tooling over custom builds

Estimated SAM: \$1.5–2.5 billion annually

Serviceable Obtainable Market (SOM)

The initial SOM targets:

- UK Financial Services institutions (50+ Tier 1–3 banks, insurers, asset managers)
- Microsoft Azure-committed organisations
- Early adopters of automated migration governance
- Phase 1: 10–20 enterprise customers

Estimated SOM (Year 1–2): \$10–30 million annually

Competitive Landscape

Category 1: Manual / In-House Solutions

Description: Custom scripts, spreadsheets, manual sampling

Strengths: Familiar, no licensing cost

Weaknesses: Not scalable, no governance, error-prone, no audit trail

Our Advantage: Automation, repeatability, governance, audit readiness

Category 2: ETL Testing Tools

Examples: QuerySurge, iCEDQ, Datagaps

Strengths: Data comparison capabilities

Weaknesses: Generic, not purpose-built for migration, limited governance

Our Advantage: Migration-specific controls, release gates, financial services focus

How They Compare:

Capability	Our Platform	ETL Testing Tools	Manual
Purpose-built for migration	[OK] Yes	[X] Generic testing	[X]
10 structured controls	[OK] Built-in	[!] Partial	[X]
Release gate governance	[OK] Built-in	[X]	[X]
Audit trail	[OK] Automated	[!] Limited	[X]
Financial Services focus	[OK] Purpose-built	[X]	[X]
Docker deployment	[OK] Ready	[X]	N/A
API-first	[OK] Built	[!] Partial	[X]

Category 3: Data Quality Platforms

Examples: Informatica, Talend, Ataccama

Strengths: Broad data quality capabilities

Weaknesses: Complex, expensive, not migration-specific, long implementation

Our Advantage: Migration-focused, lightweight deployment, faster time-to-value

Category 4: Big 4 & SI Migration Practices

Examples: Accenture Migration Factory, Deloitte Migration Services

Strengths: Domain expertise, end-to-end service

Weaknesses: High cost, manual-heavy, not repeatable across clients

Our Advantage: Automated platform, consistent methodology, lower cost

Category 5: Cloud-Native Migration Tools

Examples: Azure Migrate, AWS Migration Hub, Google Cloud Migration

Strengths: Cloud infrastructure integration

Weaknesses: Infrastructure-focused, limited data validation, no governance

Our Advantage: Data-level validation, governance, Financial Services compliance

Competitive Positioning

Our platform occupies a unique position at the intersection of:

- **RegTech** — Regulatory compliance for data migration
- **Data Validation** — Automated, comprehensive, repeatable
- **Migration Governance** — Release gates, audit, compliance
- **Azure-Native** — Purpose-built for Microsoft ecosystem

Key message: Not a generic testing tool — a purpose-built **Migration Assurance Platform** for regulated Financial Services.

Market Entry Strategy

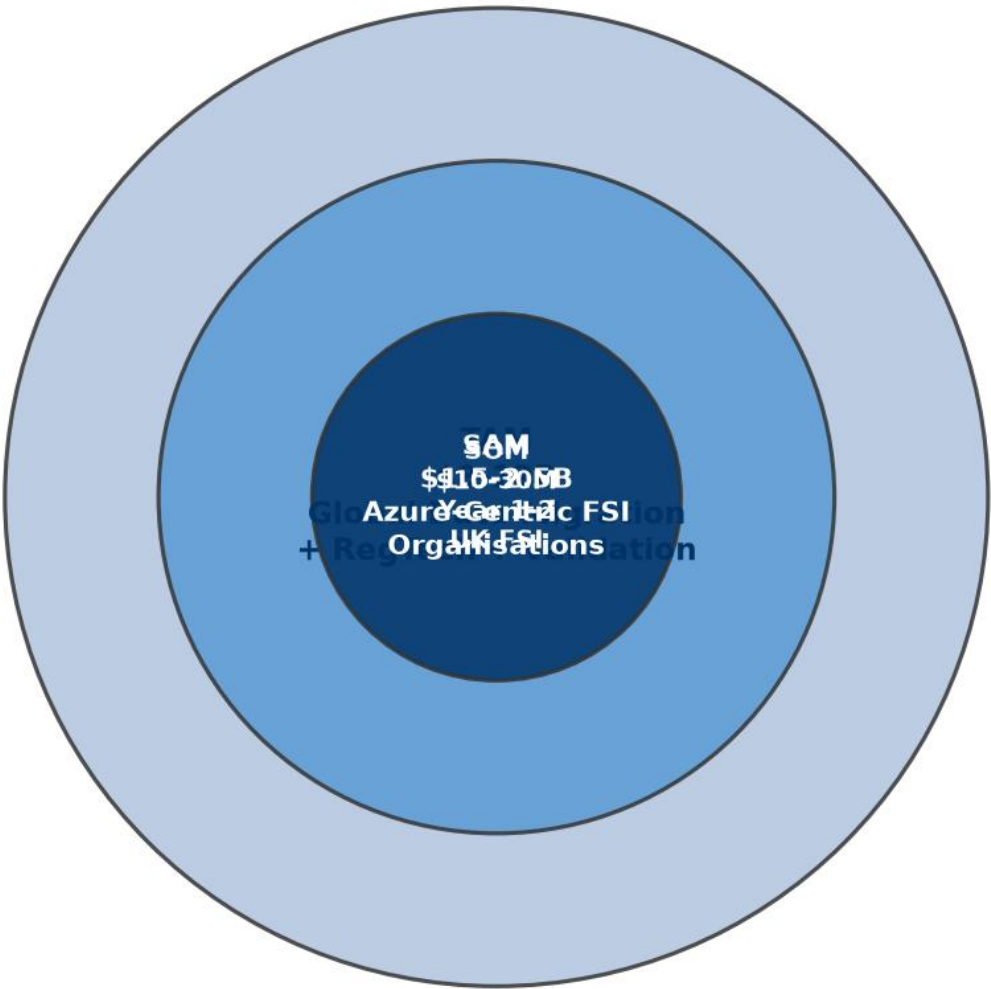
Phase	Focus	Activities
Year 1	UK Financial Services	Microsoft Founders Hub, pilot customers, direct sales
Year 2	UK + Europe	Azure Marketplace, SI partnerships, industry events
Year 3	North America	Microsoft co-sell, multi-region deployments
Year 4+	Global	Marketplace ecosystem, regulatory template library

Growth Opportunities

Opportunity	Description
Regulatory template library	Pre-built controls for Basel, SOX, FCA, PRA
AI-assisted mapping	Azure OpenAI-powered schema matching
Multi-cloud validation	Extend to AWS, GCP migrations

Dashboard & reporting	Interactive visualisation of migration quality
Managed migration service	Platform + implementation consulting
Partner ecosystem	SI embed, marketplace, co-sell

TAM / SAM / SOM



Tam Sam Som



Phase 2.4 — Business Model & Pricing Strategy

Version: 2.1

Product: FS Migration Validation Engine

Currency: GBP (£) — UK-based company

Executive Summary

The FS Migration Validation Engine follows a **SaaS subscription model** with tiered pricing based on migration project volume, number of systems, and support level. The model is designed for predictable recurring revenue (ARR) while allowing expansion through professional services and future AI modules.

Revenue is generated from four complementary streams:

- 34. **Enterprise SaaS subscriptions** — Core recurring revenue
- 35. **Professional services** — Implementation, custom controls, migration advisory
- 36. **Enterprise support & SLAs** — Premium support tiers (future)
- 37. **AI mapping module** — Azure OpenAI-powered recommendations (future)

Revenue Streams

1. SaaS Subscriptions (Primary)

Tier	Target	Price (GBP/month)	Includes
Starter	Pilot projects, small migrations	£1,500–£3,500	1 migration project, 2 databases, CLI access
Professional	Mid-size bank migrations	£3,500–£10,000	3 concurrent projects, 5 databases, API access
Enterprise	Large-scale migrations	£10,000–£35,000	Unlimited projects, multi-database, audit export, priority support
Strategic	Enterprise-wide deployment	Custom	Custom controls, multi-region, dedicated support, SLA

Typical annual contract value (ACV): £30,000–£80,000 per customer — consistent with UK enterprise regtech SaaS benchmarks.

2. Professional Services

Service	Description	Price (GBP)
Implementation	Platform setup, CI/CD integration, customer onboarding	£15,000–£40,000
Custom Controls	Development of customer-specific validation rules	£5,000–£20,000 per control
Migration Advisory	Migration validation planning and governance consulting	£1,500–£7,500/day

Professional services are typically 15–20% of total revenue in Years 1–2, reducing to 5–10% as the platform scales — consistent with SaaS industry benchmarks.

3. Future Revenue Streams

Stream	Description	Expected Launch
AI Mapping Module	Azure OpenAI-powered schema mapping recommendations	Phase 2 (Year 2)
Enterprise Support & SLA	24/7 support, guaranteed SLAs, account management	Phase 3 (Year 2–3)
Regulatory Templates	Pre-built controls for specific regulations (Basel, SOX, FCA, PRA)	Phase 3 (Year 3)

Revenue Model Rationale

The pricing model is based on:

- **Value-based pricing** — Priced on migration project value (typically £500k–£5M per programme), not per-record. A validation platform delivering 0.5–1% risk reduction on a £2M migration programme justifies £50k–£100k annual investment.
 - **Tiered progression** — Clear upgrade path from pilot (£1.5k/mo) to enterprise-scale (£35k+/mo) as customers expand usage.
 - **Subscription model** — Predictable ARR with annual contracts (typical UK enterprise procurement).
 - **Low upfront, high lifetime** — Low barrier to entry (£1.5k/mo starter), high expansion revenue as customers validate more migrations.
-

Unit Economics (Illustrative)

Metric	Year 1	Year 2	Year 3
Customers	10	40	120
Average ARR/Customer	£45,000	£60,000	£70,000
ARR	£450,000	£2,400,000	£8,400,000
Gross Margin	78%	82%	85%
CAC	£28,000	£22,000	£18,000
LTV (avg 4 years)	£180,000	£240,000	£280,000
LTV:CAC	6:1	11:1	16:1

Notes:

- CAC includes founder-led sales time (Year 1) and dedicated sales resource (Years 2–3)
- LTV:CAC ratio of 6:1 in Year 1 is healthy for enterprise SaaS; improves to 16:1 at scale
- Churn expected at <10% in Year 1, declining to <5% by Year 3 — consistent with enterprise regtech benchmarks

Azure Infrastructure Cost Model

Service	Monthly (GBP) — MVP	Monthly (GBP) — Scale
Azure Container Apps	£150–£400	£800–£2,500
Azure SQL Database	£120–£250	£400–£1,200
Azure Blob Storage	£40–£80	£150–£400
Azure API Management	£80–£150	£250–£500
Azure Key Vault	£8–£15	£40–£80
Azure Monitor	£40–£80	£150–£400
Total per customer	£438–£975	£1,790–£5,080

As the platform moves to multi-tenant SaaS, infrastructure costs per customer reduce significantly through shared tenancy.

Pricing Principles

Principle	Rationale
Value-based	Priced on migration project value, not per-record
Tiered	Clear upgrade path as customer needs grow
Subscription	Predictable recurring revenue (ARR) — essential for UK investors
Enterprise annual contracts	Standard UK enterprise procurement model — annual commitments
Low entry, high expansion	Starter tier for pilots, expansion revenue from Enterprise tier

UK Market Context

This pricing reflects UK Financial Services market conditions:

- UK banks typically spend **£500k–£5M** per migration programme
- UK enterprise regtech SaaS pricing ranges **£30k–£150k ACV**
- UK investors expect GBP-denominated projections with realistic UK salary costs
- UK corporate tax rate: 25% (2024–25)
- R&D tax credits available for software development (up to 27% of qualifying costs)
- Azure Founders Hub provides up to **£120,000 in free Azure credits** (GBP equivalent of \$150k)

Azure Marketplace Strategy

The platform will be available through:

- **Azure Marketplace** — Transactable SaaS offer (GBP pricing)
- **Microsoft Co-Sell** — Eligible for Microsoft field seller incentives
- **Private Offers** — Custom pricing for large enterprise deals

This aligns with Microsoft's partner ecosystem and provides UK Financial Services customers with Azure consumption commitment benefits.

Phase 2.5 — Go-to-Market Strategy & 3-Year Financial Projections

Version: 2.1

Product: FS Migration Validation Engine

Currency: GBP (£) — UK-based company

Go-to-Market Strategy

Target Customer Segments

Primary (Year 1–2):

Segment	Target	Entry Strategy
Tier 2–3 Banks	UK challenger banks, building societies	Direct founder-led sales, pilot projects
Insurance	UK insurers with migration programmes	Microsoft partner referrals
FinTech	Payment processors, digital banks	Azure Marketplace listing

Secondary (Year 2–3):

Segment	Target	Entry Strategy
Tier 1 Banks	Barclays, Lloyds, HSBC, NatWest	SI partnerships, longer sales cycles
Asset Management	Fund managers, wealth platforms	Industry events, case studies
Regulated Enterprises	Healthcare, Government	Compliance-led positioning

Go-to-Market Channels

Channel	Year 1 Contribution	Year 2 Contribution	Year 3 Contribution
Direct Sales (founder-led)	60%	40%	25%
Azure Marketplace	20%	25%	30%
System Integrators	10%	20%	25%
Microsoft Co-Sell	5%	10%	15%
Inbound / Content	5%	5%	5%

Sales Cycle

Segment	Cycle Length	Key Decision Makers
Small/Medium Banks	2–4 months	CTO, Head of Data, Programme Director
Large Banks	6–12 months	CIO, CRO, Head of Migration
Insurance	3–6 months	CTO, Head of Transformation
FinTech	1–3 months	CTO, VP Engineering

Customer Acquisition Strategy

Year 1:

- Secure 3–5 pilot customers in UK Financial Services
- Build case studies and reference accounts
- Establish Azure Marketplace presence
- Leverage Azure Founders Hub credits (£120k) for infrastructure

Year 2:

- Scale to 20–30 customers across UK + Europe
- Establish SI partnerships (2–3 partners — Accenture, Deloitte UK, Infosys UK)
- ISO 27001 certification

Year 3:

- 80–100 customers globally
- Microsoft co-sell programme participation
- North America market entry (with USD pricing option)

3-Year Financial Projections

Revenue Forecast

Metric	Year 1	Year 2	Year 3
Customers	10	40	120
Average ARR/Customer	£45,000	£60,000	£70,000
SaaS Revenue	£450,000	£2,400,000	£8,400,000
Professional Services	£100,000	£300,000	£500,000

Total Revenue	£550,000	£2,700,000	£8,900,000
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Revenue assumptions:

- Year 1: 10 customers at £3,750/mo average (£45k ARR) + implementation fees
- Year 2: 40 customers at £5,000/mo average (£60k ARR) — expansion revenue from existing customers via up-sell
- Year 3: 120 customers at £5,833/mo average (£70k ARR) — mix of new logos and expansion
- Professional services: 1 implementation per new customer (£15k–£30k each)
- Churn: 10% Year 1 → 8% Year 2 → 5% Year 3

Cost of Sales

Category	Year 1	Year 2	Year 3
Azure Infrastructure	£60,000	£240,000	£720,000
Customer Support (1 FTE)	£40,000	£80,000	£200,000
Third-party Licences	£20,000	£50,000	£100,000
Total CoS	£120,000	£370,000	£1,020,000
Gross Margin	78%	86%	89%

Cost assumptions:

- Azure costs reduce per-customer as shared tenancy scales
- Support: 0.5 FTE in Year 1 → 2 FTE by Year 3
- Third-party: GitHub, monitoring, security scanning tools

Operating Expenses

Category	Year 1	Year 2	Year 3
Salaries & Benefits			
Founder/CEO	£60,000	£70,000	£80,000
Senior Engineer (Python/Azure)	£90,000	£95,000	£100,000
Junior Engineer	£50,000	£55,000	£60,000
Sales/Business Development	—	£60,000	£80,000
Customer Success	—	—	£55,000
Total Salaries	*£200,000*	*£280,000*	*£375,000*

Employer NI & Pension (est. 20%)	£40,000	£56,000	£75,000
Total Staff Costs	£240,000	£336,000	£450,000
Other OpEx			
Office & Equipment	£30,000	£40,000	£60,000
Sales & Marketing	£40,000	£100,000	£200,000
Professional Advisors (legal, accounting)	£25,000	£35,000	£50,000
Insurance & Compliance	£15,000	£25,000	£40,000
Travel & Events	£10,000	£30,000	£50,000
R&D (non-staff)	£20,000	£30,000	£50,000
Total Other OpEx	£140,000	£260,000	£450,000
Total Operating Expenses	£380,000	£596,000	£900,000

Salary assumptions reflect UK market rates (outside London):

- Senior Python/Azure engineer: £85k–£100k typical for UK startup/Fintech
- Junior engineer: £45k–£55k
- Sales: £55k–£70k base + commission
- Employer NI (13.8%) + pension (3–5%) estimated at 20% on total salary

EBITDA

Year	Revenue	Total Costs (CoS + OpEx)	EBITDA
Year 1	£550,000	£500,000	£50,000
Year 2	£2,700,000	£966,000	£1,734,000
Year 3	£8,900,000	£1,920,000	£6,980,000

Year 1 reaches break-even with £50k EBITDA — achievable due to founder-led sales, low overhead, and Azure Founders Hub credits covering infrastructure.

Key Metrics

KPI	Year 1	Year 2	Year 3	Benchmark
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ARR	£450,000	£2,400,000	£8,400,000	UK regtech median: £500k→£3M→£10M+
Gross Margin	78%	86%	89%	SaaS target: >70% Year 1, >80% by Year 3
Net Revenue Retention	105%	115%	125%	Enterprise SaaS median: 110%+
Customer Churn	<10%	<8%	<5%	Enterprise target: <10% annual
CAC	£28,000	£22,000	£18,000	UK enterprise SaaS: £15k–£35k
LTV	£180,000	£240,000	£280,000	Based on 4-year avg customer life
LTV:CAC	6:1	11:1	16:1	Healthy SaaS: >3:1, Great: >10:1
EBITDA Margin	9%	64%	78%	UK SaaS Rule of 40 target
Headcount	3	5	8	Capital-efficient — typical UK startup

Funding Requirement

Target: £500,000–£750,000 Seed Round (optional — can bootstrap if Founders Hub credits sufficient)

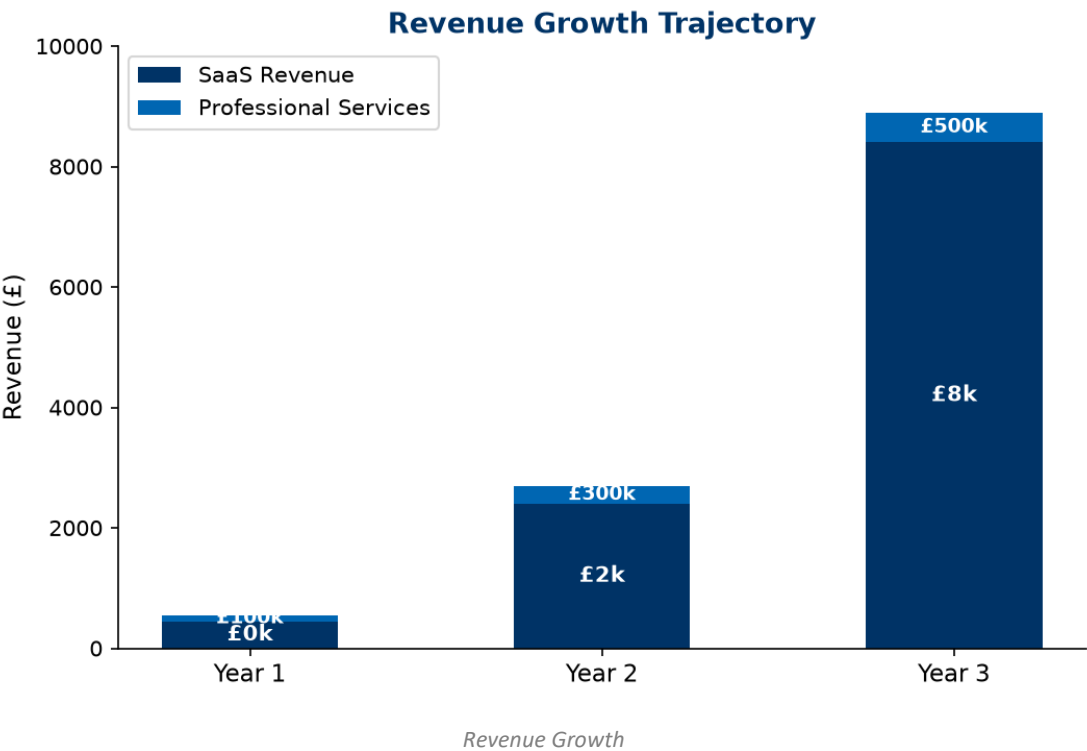
Use of Funds:

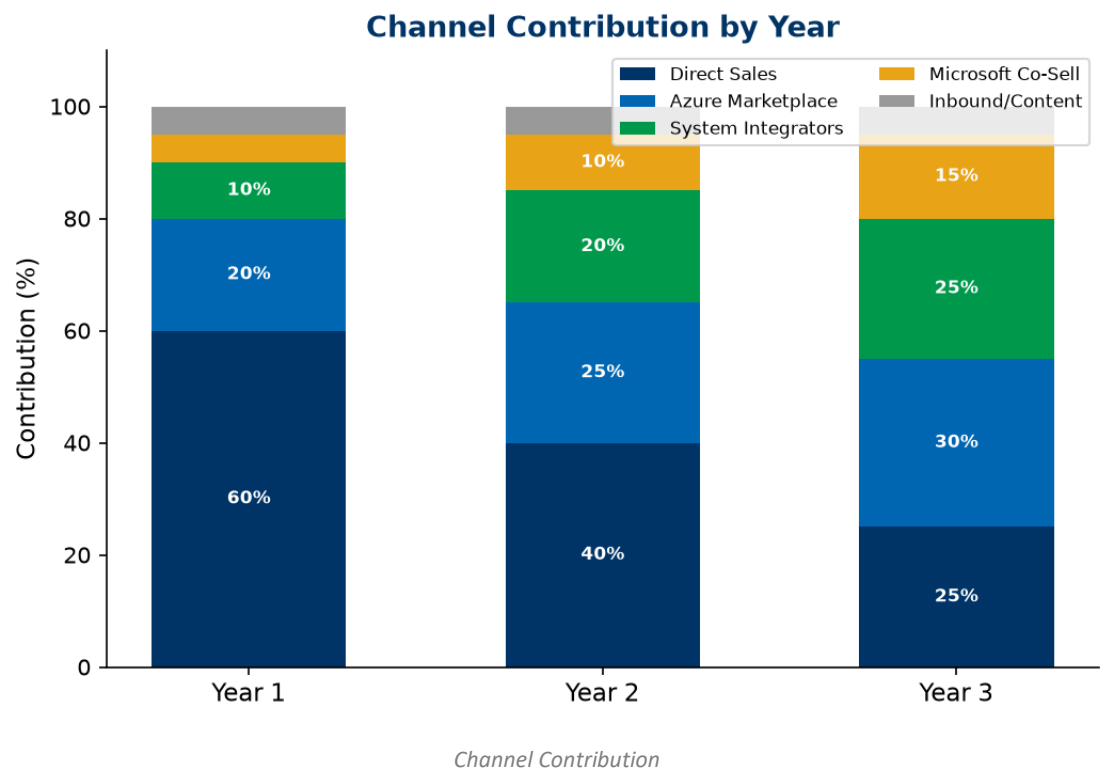
Area	Allocation	Amount (GBP)
Product Development (Phase 2)	40%	£200,000–£300,000
AI Mapping Module (Azure OpenAI)	20%	£100,000–£150,000
Customer Pilots & Validation	20%	£100,000–£150,000
Security Certifications (ISO 27001)	10%	£50,000–£75,000
Marketplace & Sales Readiness	10%	£50,000–£75,000

Alternative Path: Azure Founders Hub provides up to £120,000 in Azure credits. Combined with founder funding and early customer revenue, the business can reach breakeven without external equity in Year 1.

UK-specific funding options:

- **Azure Founders Hub** — £120k Azure credits + support
- **Innovate UK Smart Grants** — Up to £500k for R&D projects
- **UK Enterprise Investment Scheme (EIS)** — Tax relief for angel investors (30% income tax relief)
- **British Business Bank Start Up Loans** — Up to £25k per founder





Phase 2.6 — Financial Model & Funding Strategy

Version: 2.1

Product: FS Migration Validation Engine

Currency: GBP (£) — UK-based company

Financial Model Summary

3-Year P&L

Item	Year 1	Year 2	Year 3
Customers	10	40	120
SaaS Revenue	£450,000	£2,400,000	£8,400,000
Professional Services	£100,000	£300,000	£500,000
Total Revenue	£550,000	£2,700,000	£8,900,000
Cost of Sales	(£120,000)	(£370,000)	(£1,020,000)
Gross Profit	£430,000	£2,330,000	£7,880,000
OpEx (Staff + Other)	(£380,000)	(£596,000)	(£900,000)
EBITDA	£50,000	£1,734,000	£6,980,000
EBITDA Margin	9%	64%	78%

The business reaches EBITDA-positive in Year 1 — achievable through low overhead, founder-led operations, and Azure Founders Hub credits (£120k) covering infrastructure costs.

Revenue Build-Up

Component	Year 1	Year 2	Year 3	Assumption
New customers	10	30	80	Growth driven by founder-led sales + SI partnerships
Retained customers	—	9	37	90% retention Year 1 → 92% Year 2
Total customers (EoY)	10	40	120	Net growth after churn
Average	£45,000	£60,000	£70,000	Expansion via up-

ARR/customer				sell + new logos
SaaS ARR	£450,000	£2,400,000	£8,400,000	Recurring subscription revenue
Professional services	£100,000	£300,000	£500,000	Implementation + custom controls

Azure Infrastructure Cost Model

Service	Monthly (MVP)	Monthly (Scale)
Azure Container Apps	£150–£400	£800–£2,500
Azure SQL Database	£120–£250	£400–£1,200
Azure Blob Storage	£40–£80	£150–£400
Azure API Management	£80–£150	£250–£500
Azure Key Vault	£8–£15	£40–£80
Azure Monitor	£40–£80	£150–£400
Monthly Total	£438–£975	£1,790–£5,080
Annual Total per customer	£5,256–£11,700	£21,480–£60,960

Year 1: Single-tenant deployment costs ~£5k/customer/year.

Year 3: Multi-tenant SaaS model reduces per-customer costs to £1k–£2k/customer/year.

Hiring Plan

Year	Role	Salary (GBP)	Notes
Y1	Founder/CEO	£60,000	UK founder salary, below market
Y1	Senior Engineer (Python/Azure)	£90,000	UK market rate for senior engineer
Y1	Junior Engineer	£50,000	UK graduate + 1–2 years experience
Y2	Sales/Business Development	£60,000 + commission	Target for 30 new customers
Y3	Customer Success	£55,000	Support 120 customer accounts

Y3	Additional Engineer	£65,000	Scale development capacity
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All salaries based on UK market rates (outside London). Employer NI (13.8%) + pension (3–5%) = ~20% on top.

Funding Strategy

Azure Founders Hub (Primary — Year 1)

Benefit	Value (GBP)
Free Azure credits	Up to £120,000
Microsoft partner enrolment	Free
Azure Marketplace listing	Included
Technical support	Included
Go-to-market support	Included

The Founders Hub credits alone cover Year 1 Azure infrastructure (£60k), with remaining credits available for development and testing environments.

Seed Round (If Required)

Item	Detail
Target	£500,000–£750,000
Valuation basis	UK angel/Seed typical: £1.5M–£3M pre-money for SaaS with working prototype
Use	Product development, AI module, customer pilots, security certifications
Timing	End of Year 1, after Founders Hub credits utilised
Investor target	UK angel networks, EIS-qualified investors

Future Funding

Round	Target	Timing	Trigger
Seed	£500k–£750k	Year 1–2	Product validation, 10+ customers, £450k ARR

Series A	£3M–£5M	Year 3	£2M+ ARR, proven GTM, SI partnerships
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UK-Specific Funding Options

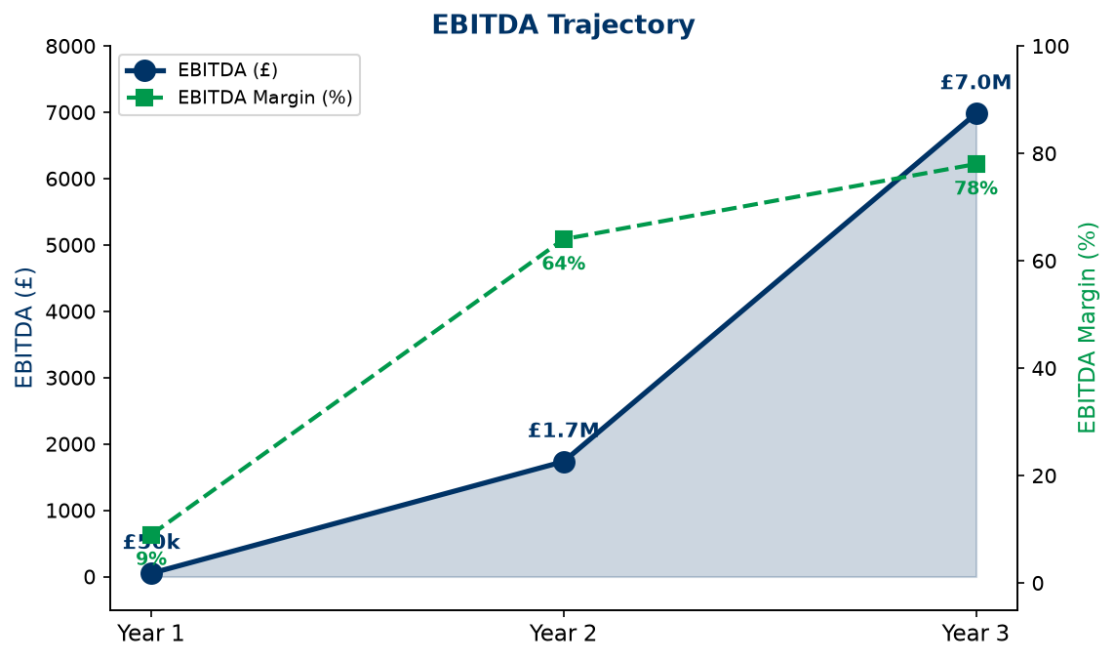
Source	Amount	Eligibility
Azure Founders Hub	Up to £120k credits	Startup stage
Innovate UK Smart Grant	Up to £500k	R&D projects, competitive application
EIS (Enterprise Investment Scheme)	Tax relief for investors	Must be EIS-qualified company
UK SEIS	Up to £250k investment	Early-stage, less than 2 years old
British Business Bank	Various	Startup loans, growth guarantee

Sensitivity Analysis

Scenario: Slower Customer Acquisition

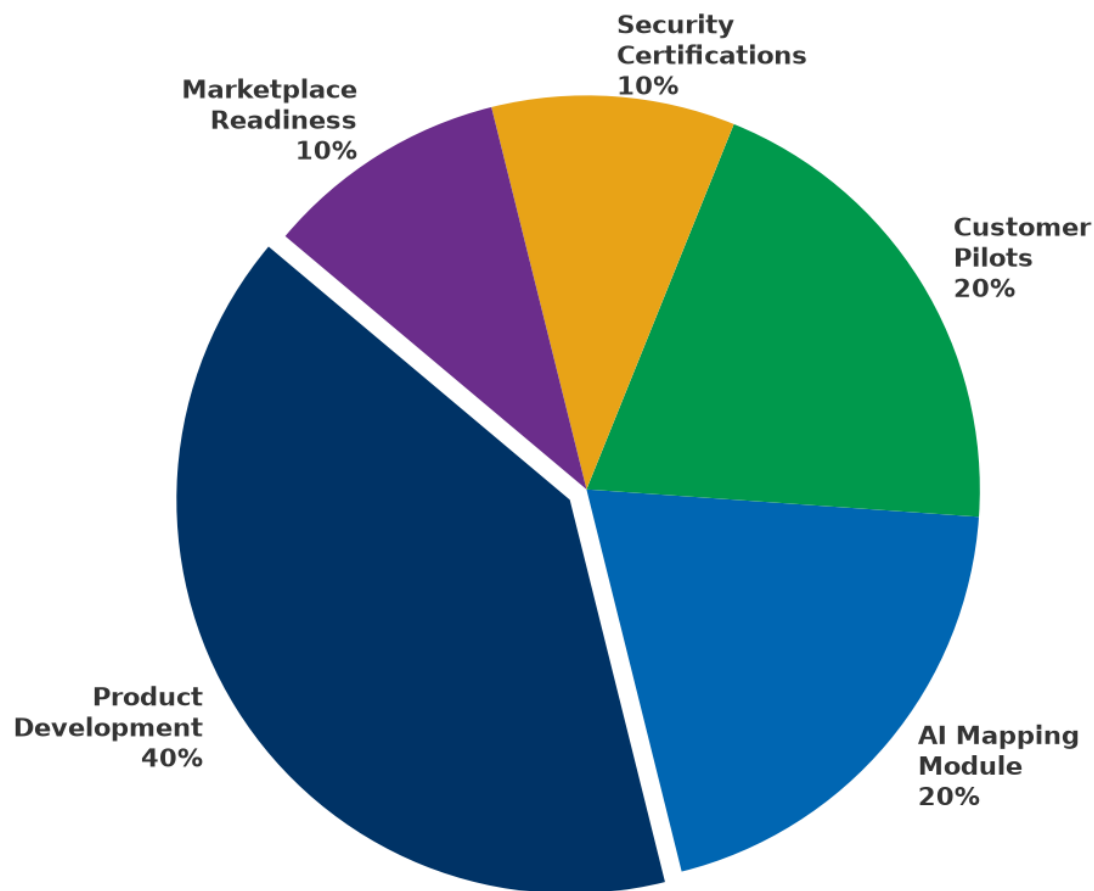
Metric	Base Case	Downside (-30%)	Upside (+30%)
Year 1 customers	10	7	13
Year 1 revenue	£550,000	£385,000	£715,000
Year 1 EBITDA	£50,000	(£115,000)	£215,000
Year 3 ARR	£8.4M	£5.9M	£10.9M
Breakeven	Year 1	Year 2 (Q3)	Year 1

The business model is resilient — even at 30% lower customer acquisition, breakeven is achieved by mid-Year 2 due to low fixed costs and variable infrastructure spend.



Ebitda Trajectory

Use of Funds (£500k-£750k Seed)



Use Of Funds

Phase 2.7 — Competitive Differentiation

Version: 2.0

Product: FS Migration Validation Engine

Competitive Landscape

Category	Typical Vendors	Strengths	Weaknesses	Our Advantage
Manual / In-House	Custom scripts, spreadsheets	Familiar, no license cost	Not scalable, no governance	Automation, repeatability, audit
ETL Testing Tools	QuerySurge, iCEDQ, Datagaps	Data comparison	Not migration-focused, limited governance	Migration-specific controls, release gates
Data Quality Platforms	Informatica, Talend, Ataccama	Broad data quality	Complex, expensive, long implementation	Lightweight, migration-focused, faster deployment
Big 4 / SI Services	Accenture, Deloitte Migration Services	Domain expertise	High cost, manual-heavy, not repeatable	Automated platform, consistent methodology
Cloud Migration Tools	Azure Migrate, AWS Migration Hub	Infrastructure integration	Infrastructure-focused, limited data validation	Data-level validation, governance, compliance

Differentiation Matrix

Capability	Our Platform	ETL Testing	Data Quality	Manual
Purpose-built for migration	[OK]	[X]	[X]	[X]
10 structured controls	[OK]	[!]	[!]	[X]
Release gate governance	[OK]	[X]	[X]	[X]
Audit trail (automated)	[OK]	[!]	[!]	[X]

Financial Services focus	[OK]	[X]	[X]	[X]
Docker deployment	[OK]	[X]	[!]	N/A
API-first	[OK]	[!]	[!]	[X]
Objective scoring	[OK]	[!]	[!]	[X]

Core Differentiators

1. Purpose-Built for Migration

Not a general-purpose testing or data quality tool — designed specifically for data migration validation with 10 structured controls that cover the complete migration lifecycle.

2. Governance-First Design

Release gates, audit trails, exception registers, and compliance reporting are built into every execution — not added as afterthoughts.

3. Financial Services Focus

Purpose-built for regulated environments. Controls align with regulatory requirements (FCA, PRA, Basel, SOX).

4. Lightweight Deployment

Docker-native, deploy in minutes. No complex installation, no heavy infrastructure requirements.

5. API-First Architecture

Integrate with existing CI/CD pipelines, deployment workflows, and enterprise toolchains.

6. Objective, Repeatable Scoring

Automated scoring removes subjectivity. Same inputs always produce same outputs — essential for regulated environments.

Why Customers Choose Our Platform

Customer Need	Manual Approach	Our Platform
Complete validation coverage	Sampling only (~20%)	100% of records
Regulatory audit evidence	Manual evidence gathering	Automated CSV export
Repeatable process	New scripts each migration	Standardised framework

Migration governance	Spreadsheet tracking	Release gates + audit
Objective quality metrics	Subjective assessment	Automated scoring
Fast deployment	Weeks of setup	Docker, minutes to deploy

Phase 2.8 — Product Roadmap

Version: 2.0

Product: FS Migration Validation Engine

Vision

To become the **standard platform for regulated data migration assurance** — helping Financial Services institutions validate every data migration with automated controls, objective scoring, and auditable governance.

Product Evolution



Year 1 — Foundation & Validation (Current)

Objective

Establish the platform with Financial Services customers and validate product-market fit.

Completed

Capability	Status
CLI Engine (10 controls)	[OK] Complete
PostgreSQL schema	[OK] Complete
Scoring engine	[OK] Complete
Release gates	[OK] Complete
Audit export (CSV)	[OK] Complete

Docker deployment	[OK] Complete
Control dependency DAG	[OK] Complete
Security scanning	[OK] Complete
REST API (FastAPI)	[OK] Built
Schema discovery	[OK] Built
Rule executor	[OK] Built

In Development

Capability	Timeline
Web dashboard (interactive)	Q3
AI-assisted mapping (Azure OpenAI)	Q4
Enterprise SSO (Entra ID)	Q4

Milestones

- [OK] 5 pilot customers in UK Financial Services
- [OK] Azure Marketplace listing
- ☐ ISO 27001 certification (Q4)
- ☐ First paid enterprise customer (Q3)

Year 2 — Enterprise Platform

Objective

Expand to multi-tenant SaaS with enterprise-grade capabilities.

Capability	Description	Priority
Multi-tenant architecture	Isolated customer workspaces	High
Interactive dashboard	Real-time migration quality visualisation	High
AI mapping recommendations	Azure OpenAI-powered schema matching	High
User management	RBAC, team collaboration	Medium
Advanced reporting	Custom report builder	Medium

API versioning	v2 API with enhanced capabilities	Medium
Power BI integration	Embedded analytics	Low

Milestones

- 30+ enterprise customers
- European market expansion
- 2+ system integrator partnerships
- Microsoft co-sell programme enrolment

Year 3 — AI-Powered Migration Intelligence

Objective

Transition from automated validation to intelligent migration decision support.

Capability	Description
Intelligent mapping engine	ML-powered schema matching with confidence scoring
Predictive risk scoring	AI models predict migration failure risk
Regulatory template library	Pre-built controls for Basel, SOX, FCA, PRA
Anomaly detection	AI identifies unusual data patterns during migration
Natural language queries	Ask questions about migration quality in plain English
Automated remediation suggestions	AI recommends fixes for failed controls

Milestones

- 100+ customers
- \$10M+ ARR
- North America market entry
- Series A fundraising

Year 4+ — Intelligent Migration Platform

Objective

Become the comprehensive platform for migration governance and intelligence.

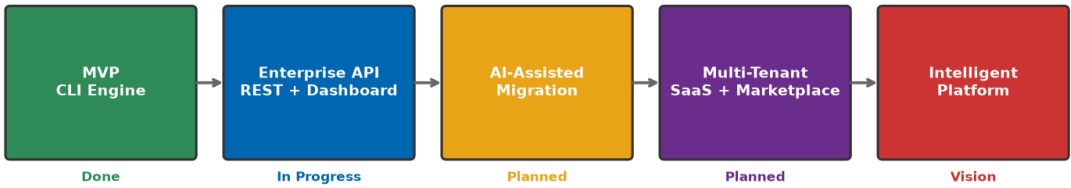
Capability	Description
------------	-------------

AI agents	Autonomous migration monitoring and alerting
Microsoft Fabric integration	Enterprise data platform connectivity
Marketplace	Third-party control and integration marketplace
Multi-cloud	AWS, GCP migration validation support
Compliance automation	Automated regulatory reporting for migrations
Migration knowledge graph	Cross-project migration intelligence

Technology Roadmap

Capability	Y1	Y2	Y3	Y4+
CLI Engine	[OK]			
REST API	[OK]			
PostgreSQL	[OK]			
Docker	[OK]			
Web Dashboard		[OK]		
Azure OpenAI		[OK]		
Multi-Tenant SaaS		[OK]		
RBAC / SSO		[OK]		
Power BI		[OK]		
AI Mapping Engine			[OK]	
Regulatory Templates			[OK]	
Predictive Risk			[OK]	
Microsoft Fabric				[OK]
Multi-Cloud				[OK]
AI Agents				[OK]
Marketplace				[OK]

Product Roadmap



Product Roadmap

Phase 2.9 — Investor FAQ

Version: 2.1

Product: FS Migration Validation Engine

Currency: GBP (£) — UK-based company

Company & Vision

1. What problem are you solving?

Financial Services institutions migrating data between systems lack automated validation tools. They rely on manual sampling, spreadsheets, and custom scripts — resulting in incomplete coverage, no audit trail, and high risk of undetected data errors. In regulated environments, this is unacceptable.

2. Why is this problem important now?

Three trends converge: (1) accelerating cloud migration in Financial Services, (2) increasing regulatory pressure for data integrity (FCA, PRA, Basel IV, SOX), and (3) retirement of legacy platform experts. Institutions need automated, governed, and audit-ready validation — not manual processes from a bygone era.

3. Who are your customers?

Tier 1–3 banks, insurance companies, asset managers, FinTechs, and regulated enterprises undergoing data migration. Initial focus on UK Financial Services, expanding to Europe and North America.

4. How large is the opportunity?

Global TAM estimated at \$8–12B annually across Financial Services data migration, RegTech, and data validation. SAM of \$1.5–2.5B for Azure-centric FSI organisations. SOM of \$10–30M in Year 1–2.

Product

5. What does the platform do?

Automated data migration validation. It discovers source/target schemas, runs 10 structured validation controls (row counts, data types, keys, business rules, etc.), scores migration quality, enforces release gates, and produces audit-ready CSV exports.

6. What makes you different from ETL testing tools?

We're purpose-built for **data migration**, not generic ETL testing. Our 10 controls are standardised for migration validation. We include governance features (release gates, audit trails, dependency management) that ETL tools don't have. And we're focused specifically on regulated Financial Services.

7. Is your technology defensible?

Yes — our defensibility comes from: (1) domain expertise in Financial Services migration, (2) proprietary control framework (C01–C010), (3) modular Platform Core architecture, (4) customer implementation knowledge, and (5) regulatory alignment that generic tools can't easily replicate.

8. Do you use AI?

Our current platform is rule-based and metadata-driven — delivering reliable, deterministic validation. We plan to integrate Azure OpenAI for AI-assisted mapping recommendations in Phase 2. We prioritise accuracy and auditability over black-box AI.

Market

9. Who are your competitors?

Manual in-house solutions, ETL testing tools (QuerySurge, iCEDQ), data quality platforms (Informatica, Talend), Big 4 migration services (Accenture, Deloitte), and cloud migration tools (Azure Migrate). None offer our combination of migration-focused controls, governance, audit, and lightweight deployment.

10. Why will customers buy from you instead of building internally?

Building an equivalent platform requires expertise in migration validation, database internals, governance frameworks, and regulatory compliance — plus months of development. Our platform is ready to deploy today, standardised, and continuously improved.

11. What is your pricing model?

SaaS subscription — £1,500–£3,500/month (Starter), £3,500–£10,000/month (Professional), £10,000–£35,000/month (Enterprise), custom for Strategic deployments. Plus professional services for implementation (£15k–£40k) and custom controls (£5k–£20k each).

Business

12. What is your business model?

Recurring SaaS subscriptions with tiered pricing based on migration project volume and support level. Additional revenue from professional services and future AI modules.

13. What are your expected margins?

Gross margins of 80%+ as we scale, typical for SaaS platforms. Operating margins improve from Year 2 as development costs stabilise and revenue scales.

14. How will you acquire customers?

Founder-led direct sales to Financial Services CTOs and Heads of Data, Azure Marketplace listings, system integrator partnerships (Accenture, Deloitte, Infosys), and Microsoft co-sell programme.

15. What is your sales cycle?

2–4 months for mid-size banks and FinTech, 6–12 months for large Tier 1 banks. Pilot projects provide a faster entry path.

Funding**16. How much funding are you seeking?**

£500,000–£750,000 seed round, or utilisation of Azure Founders Hub credits (up to £120,000) for initial infrastructure and development. As a UK-based company, we are also exploring Innovate UK Smart Grants and EIS-qualified angel investment.

17. How will the funds be used?

40% product development (Phase 2), 20% AI mapping module (Azure OpenAI), 20% customer pilots, 10% security certifications, 10% marketplace readiness.

18. What milestones will this funding achieve?

Product validation with 5+ Financial Services customers, Azure Marketplace listing, ISO 27001 certification, repeatable sales process, and £450k+ ARR.

Risks**19. What are the biggest risks?**

Enterprise sales cycles are longer than consumer SaaS. Competition from established data quality vendors. Customer adoption of new validation methodologies. These are mitigated through pilot programmes, SI partnerships, and Financial Services domain focus.

20. What prevents Microsoft or a large vendor from copying you?

Large vendors move slowly in specialised niches. Our Financial Services domain expertise, migration-specific control framework, and customer implementation knowledge create a moat. We're not competing with Azure Migrate — we complement it.

Phase 2.10 — Demo Script

Version: 2.0

Product: FS Migration Validation Engine

Duration: 10–15 minutes

Objective

Demonstrate how the FS Migration Validation Engine automates data migration validation for Financial Services — from schema discovery through 10 validation controls, objective scoring, release gate enforcement, and audit-ready exports.

Demo Flow

Section	Duration
Introduction & Problem	1 min
Platform Overview	2 mins
Schema Discovery	2 mins
Validation Execution (C01–C010)	3 mins
Scoring & Release Gates	2 mins
Audit Export	2 mins
API Integration	1 min
Closing & Business Value	2 mins
Total	~15 minutes

1. Opening (1 Minute)

"Financial Services institutions migrating data between systems face a critical challenge: how to ensure accuracy, completeness, and regulatory compliance when millions of records are in motion.

Traditional approaches rely on manual sampling, spreadsheets, and custom scripts — which are slow, error-prone, and fail regulatory scrutiny.

The FS Migration Validation Engine automates this process — providing complete validation coverage, objective scoring, governance, and audit-ready evidence."

2. Platform Overview (2 Minutes)

Show the CLI interface and explain the architecture:

```
FS Migration Validation Engine v1.4
├─ 10 Migration Controls (C01-C010)
├─ Automated Scoring Engine
├─ Release Gate Enforcement
├─ Audit Export (CSV)
├─ REST API (FastAPI)
└─ Docker Deployment
```

"The engine runs as a CLI tool, REST API, or Docker container. It connects to source and target databases, runs 10 standardised validation controls, scores the results, enforces release gates, and produces audit-ready CSV exports."

3. Schema Discovery (2 Minutes)

Show the discovery command:

```
python app/main.py discover --config config.yaml
```

Highlight:

- Automated connection to source and target databases
- Schema metadata extraction
- Table, column, and relationship discovery

"The discovery engine automatically connects to source and target databases, extracts schema information, and builds a metadata repository. No manual schema documentation required."

4. Validation Execution (3 Minutes)

Show the run command:

```
python app/main.py run --config config.yaml
```

Walk through controls being executed:

Control	Description	Example Output
---------	-------------	----------------

C01	Row count reconciliation	"PASS — Source: 1,234,567 vs Target: 1,234,567"
C02	Data type validation	"PASS — All column types match"
C03	Nullability check	"PASS — All constraints satisfied"
C04	Primary key integrity	"PASS — No duplicates"
C05	Foreign key integrity	"PASS — All references valid"
C06	Business rule validation	"3 WARNINGS — See exception register"
C07–C010	Additional controls	"PASS / WARN / FAIL per control"

"Each control runs independently with failure isolation. Results are captured with unique batch IDs for full traceability. The control dependency DAG ensures correct execution order — C02 waits for C01, C09 waits for C03."

5. Scoring & Release Gates (2 Minutes)

Show the scoring output:

```
Batch: ae40b96c-20da-4972-bb29-bff3c2451ae0
Overall Score: 87/100
Controls Passed: 8/10
Controls Warning: 2/10
Controls Failed: 0/10

RELEASE GATE: PASSED (minimum 80% threshold met)
Migration: APPROVED
```

"The scoring engine produces objective, repeatable quality metrics. The release gate enforces a minimum quality threshold — if the score drops below 80%, the migration is blocked."

6. Audit Export (2 Minutes)

Show the export command:

```
python app/main.py export --config config.yaml --batch-id <id>
```

Show sample CSV output:

```
batch_id,control_id,status,score,exceptions,timestamp
ae40...,C01,PASS,100,0,2026-06-23 10:00:00
ae40...,C02,PASS,100,0,2026-06-23 10:00:05
ae40...,C06,WARN,70,3,2026-06-23 10:00:30
```


*"Every validation execution produces a complete audit trail — suitable for regulatory review.
No manual evidence gathering required."*

7. API Integration (1 Minute)

Show a sample API call:

```
curl -X POST https://api.example.com/api/v1/validate \  
-H "Content-Type: application/json" \  
-d '{"project_id": "ae40...", "controls": ["C01", "C02"]}'
```

*"The FastAPI REST API enables integration with existing CI/CD pipelines, deployment workflows,
and enterprise toolchains."*

8. Closing Summary (2 Minutes)

"In just a few minutes, we've demonstrated:

*[OK] **Automated schema discovery** — No manual analysis required*

*[OK] **10 structured validation controls** — Complete, standardised coverage*

*[OK] **Objective scoring** — Repeatable quality metrics*

*[OK] **Release gate governance** — Automated migration approval/blocking*

*[OK] **Audit-ready exports** — CSV evidence for regulatory review*

*The platform is Azure-native, Docker-deployable, and purpose-built for regulated Financial
Services data migration.*

*Our mission is to transform data migration validation from a manual, high-risk exercise into an
automated, governed, and audit-ready process."*

Demo Environment Requirements

Item	Details
Engine	Running instance of FS Migration Validation Engine
Databases	Source and target PostgreSQL databases with demo data

Configuration	`config.yaml` with DemoBank settings
CLI	Terminal with Python environment activated
Sample Data	`sql/demo/` — demo source and target data scripts
Backup	`engine_backup.sql`, `source_backup.sql`, `target_backup.sql`

Key Messages to Reinforce

- 38. **Purpose-built for Financial Services** — Not a generic ETL tool
- 39. **Automated and repeatable** — Same inputs, same outputs, every time
- 40. **Governance-first** — Release gates, audit trails, compliance built-in
- 41. **Lightweight deployment** — Docker-native, deploy in minutes
- 42. **Azure-native** — Built for the Microsoft ecosystem
- 43. **Enterprise-ready** — Secure, scalable, auditable